LOCAL STRUCTURE OF AGRAWAL'S CONGRUENCE AT $r = 5$: CYCLOTOMIC MOMENTS, THE GOLDEN UNIT, AND KUMMER DEFECTS

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ABSTRACT. Agrawal's conjecture asserts that if r is prime, $r \nmid n$, and $(X-1)^n \equiv X^n - 1 \pmod{n,\, X^r - 1}$, then n is prime or $n^2 \equiv 1 \pmod{r}$. We develop a local theory of this congruence at $r = 5$. Writing $S(p,r)$ for the set of exponents that satisfy the congruence in $\mathbb{F}_p[X]/(\Phi_r)$, we attach to each pair (p,r) a family of *cyclotomic moments* in $\mathbb{F}_r$ whose non-vanishing rigidifies the image of $S(p,r)$ modulo r. For $r = 5$ the second moment is, up to an invertible constant, the quintic character of the golden ratio: $M_2 = 3 \cdot \mathrm{ind}_5\big(\frac{1+\sqrt{5}}{2}\big)$. As a consequence, a nontrivial local image requires 5 and the golden unit to be simultaneous 5^s-th power residues for every available level s, together with an explicit 2-adic alignment; every local exception satisfies $p \equiv 1 \pmod{20}$, and the conditional splitting density is 1/25 per floor of an explicit non-abelian Kummer tower. For squarefree n we prove a mod-5 structure theorem for counterexamples, derive an explicit Korselt-type system for the minimal three-factor case, determine the exact order decomposition $T_p = 2^{b+1} \cdot 5 \cdot d_+ \cdot d_5$ of $\zeta_5 - 1$ in $\mathbb{F}_{p^4}$, and identify its tail defect as a Kummer defect: $q_p = \prod_\ell \ell^{e_\ell}$, where ℓ^{e_ℓ} measures $\zeta_5 - 1$ being an ℓ^{e_ℓ}-th power; the (-1)-side carries no defect of its own ($d_5 = \mathrm{ord}_p(5)_{\mathrm{odd}}$ exactly), so T_p is completely determined by the Kummer defects of the canonical pair $(5,\ \zeta_5 - 1)$. This exhibits the $(+1)$-side conditions of the Lenstra–Pomerance heuristic as consequences, rather than assumptions, of the local algebra; more precisely, the two coprime odd components d_5, d_+ of the order decomposition realize the two sides $p \mp 1$ of the Lenstra–Pomerance mechanism, which the local structure thus generates internally. Under the explicit local Hypothesis H4, the squarefree problem reduces exactly to the emptiness of finite resultant fibers over compatible tuples. We also replace the norm surrogate in H4 by an exact four-coefficient primitive support in the integral basis of $\mathbb{Q}(\sqrt{5})$, and exclude universally the four infinite candidate families $p - 1 = 2^b q^e$ for $b \in \{3,4,5,6\}$. We certify a stated three-factor region and seven individual empty fibers. For every fixed number s of prime factors, the terminal row gives the unconditional quartic-skeleton bound

$$M_s^{(4)}(X) \ll_s \frac{X}{\log X}(\log\log X)^{s-2},$$

saving one factor of $\log\log X$ over the unrestricted almost-prime count. The two remaining walls are H4 and the global emptiness of the union of the finite fibers.

Preprint v25 of July 30, 2026. Mathematical development, paper, and formalization were produced within the UNICO/NOUS pipeline, initiated and operated by Daniele Cappello, who curated the artifacts and is responsible for their provenance, stated scope, and public release. This role is not presented as specialist mathematical verification. The pipeline used GPT-5.6-Sol (xhigh) in Codex and Claude Opus 4.8/5 in Claude Code. Model-based adversarial review was used during development; no human specialist review is claimed. A kernel-pure Lean 4/Mathlib layer, with no `sorry`, covers the elementary core identified in the text. Separate Mathlib-only comparator challenges deliberately contain statement holes and are not implementation proofs. All computational claims come with reproducible scripts, hashes and stated domains.

1. Introduction

Let r be a prime and $n \geq 2$ an integer with $r \nmid n$. Agrawal's conjecture; formulated by Bhattacharjee and Pandey [2] and discussed by Agrawal in connection with the AKS primality test [1]; asserts:

Conjecture 1.1 (Agrawal). *If $(X-1)^n \equiv X^n - 1 \pmod{n, X^r - 1}$, then n is prime or $n^2 \equiv 1 \pmod r$.*

A proof would lower the complexity of AKS-type primality proving to $\widetilde{O}(\log^3 n)$. Lenstra and Pomerance [10] gave a heuristic construction suggesting that counterexamples exist and are abundant, though astronomically large: products $n = p_1 \cdots p_k$ with k odd, $p_i \equiv 3 \pmod{80}$, $p_i - 1 \mid n - 1$ and $p_i + 1 \mid n + 1$ satisfy the congruence at $r = 5$ with $n^2 \not\equiv 1 \pmod 5$. Váňa [11] gave an alternative proof of the Lenstra–Pomerance proposition through the multiplicative order of $X - 1$ in $\mathbb{Z}_n[X]/(X^5 - 1)$ and reduced the congruence to a CRT-combinable system of Korselt-type congruences; Popovych [12] generalized Lenstra's proposition; Hegde and Devaraj [14] constructed further heuristic families. No counterexample is known; distributed searches found none in the ranges attempted. The paired Carmichael/Lucas–Carmichael conditions themselves predate this conjecture: Williams posed the analogous existence problem in 1977 [5]; McIntosh recorded the associated cross-support compatibility [7]; Leng derived closely related congruence consequences in the setting of combined Fermat–Lucas tests [8]; and McIntosh–Mitra Dipra later developed finiteness and search results for the corresponding Williams numbers [6].

This paper is a study of the *local* obstruction: what happens at a single prime p, in the ring $A = \mathbb{F}_p[X]/(\Phi_r)$. Our findings can be summarized in one sentence:

> *the local degeneracies of Agrawal's congruence classified so far are governed by Kummer defects of canonical cyclotomic elements*; for $r = 5$, by the quintic characters of 5 and of the golden ratio, and, in the tail of the three-factor system, by the power-residue defect of $\zeta_5 - 1$ itself. (The scope qualifier is deliberate: the possible order-4 local branch, Hypothesis H4 below, is not yet classified.)

Main results. Write $S(p,r) = \{m \geq 1 : (X-1)^m \equiv X^m - 1 \text{ in } \mathbb{F}_p[X]/(\Phi_r)\}$ and let $\operatorname{im}_r S \subseteq (\mathbb{Z}/r)^\times$ be its image modulo r.

(1) *Moments* (§3). We attach to (p,r) moments $M_j \in \mathbb{F}_r$ built from discrete logarithms of $\zeta^a - 1$ and prove $\operatorname{im}_r S \subseteq T$, where T is the subgroup cut out by the nonzero M_j. The M_j obey exact structure theorems (Frobenius, parity, head, tail): in particular $M_0 = \log_\gamma(r) \bmod r$, so $M_0 = 0$ iff r is an r-th power residue.

(2) *The golden unit* (§4). For $r = 5$, $p \equiv 1 \pmod 5$: $M_2 = 3\,\mathrm{ind}_5(\varepsilon)$ with $\varepsilon = (1+\sqrt{5})/2$. Hence nontrivial local behaviour requires 5 *and* ε to be quintic residues. The identity lifts 5-adically: $M_0^{(s)} = \mathrm{ind}_{5^s}(5)$, $M_2^{(s)} = -2\,\mathrm{ind}_{5^s}(\varepsilon)$, forcing both keys to open *every* floor of the tower. The governing tower $L_s = \mathbb{Q}(\zeta_{5^s}, 5^{1/5^s}, \varepsilon^{1/5^s})$ satisfies $[L_s : B_s] = 25$ at every floor, whence a conditional Chebotarev density of $1/25$ per floor.

(3) *Two locks and the local classification* (§5). For $p \equiv 1 \pmod 5$: $\operatorname{im}_5 S \neq \{1\}$ iff the golden lock ($5, \varepsilon \in (\mathbb{F}_p^\times)^{5^s}$, $s = v_5(p-1)$) and the binary lock ($\nu_1 = \nu_2 \leq v_2(p-1) - 2$ for explicit 2-adic valuations of discrete logarithms) are both open. Every local exception satisfies $p \equiv 1 \pmod{20}$. All 406 known local exceptions are predicted by this classification.

(4) *Structure of squarefree counterexamples* (§6). A composite squarefree n satisfying the congruence with $n^2 \not\equiv 1 \pmod 5$ must have an odd number $k \geq 3$ of prime factors if no order-4 residue occurs in the ± 1 branches (Hypothesis H4, consistent with the stated finite censuses but still open); unconditionally, an even k forces an order-4 residue. Under H4 all factors are $\equiv \pm 2 \pmod 5$. In the quartic branch we prove the rigidity $S(p, 5) = \langle p \rangle$ unconditionally, which makes the three-factor system fully explicit: Korselt-type congruences $n/p_i \equiv p_i^{j_i} \pmod{M_i}$ with determined exponents $j_i \in \{0, 2\}$.

(5) *The order of* $\zeta - 1$ *and the Kummer defect* (§7–8). In the quartic branch, $\mathrm{N}(\zeta - 1) = 5$ gives $T_p \mid 10(p^2 - 1) \cdot$ (vertical part), with the exact decomposition $T_p = 2^{b_p+1} \cdot 5 \cdot d_{p,+} \cdot d_{p,5}$, $b_p = v_2(p^2 - 1)$, and the exact binary law $v_2(T_p) = 1 + v_2(p^2 - 1)$. The tail quotient $q_p = (p+1)_{\mathrm{odd}}/d_{p,+}$ satisfies $q_p = \prod_\ell \ell^{e_\ell}$ where e_ℓ is the Kummer defect of $\zeta - 1$ at ℓ: the deviation from the full Lenstra–Pomerance $(+1)$-side condition is *exactly* a power-residue phenomenon.

(6) *Derived Lenstra–Pomerance mechanism* (§9). After the binary filter, the three-factor system collapses to the cyclic tail system $d_{i,+} \mid p_j p_k - 1$; a weakened form of $p_i + 1 \mid n + 1$. The $(+1)$-side of the Lenstra–Pomerance construction is thereby *derived* from the local algebra as a necessary condition on the minimal skeleton, rather than imposed as a sufficient device.

(7) *Exact reduction and certified computations* (§10–11). Under H4 the squarefree problem is equivalent to the emptiness of explicit finite fibers over compatible tuples. An exact finite computation excludes the region $7 \leq p_1 < p_2 < 30{,}000$, $p_3 < 10^9$; the separate boundary $p_1 \in \{2, 3\}$ is certified for $p_3 < 2 \cdot 10^8$. The pair $(13, 37, 30517)$ shows that two complete rows can hold simultaneously, so this regional statement is not a pairwise-incompatibility theorem. Seven individual fibers are empty without any upper bound on the last prime.

(8) *Counting* (§12). The last row yields, for every fixed $s \geq 3$, the unconditional quartic-skeleton estimate

$$M_s^{(4)}(X) \ll_s \frac{X}{\log X} (\log \log X)^{s-2}.$$

Under H4 this bounds every squarefree counterexample with s factors. Separately, under Dedekind–GRH for the explicit joint Kummer composita, the local H4 profiles satisfy $N_H(X) \ll X/(\log X)^2$.

Relation to earlier work. Several ingredients of this paper are classical, and we state plainly what is not ours. The focus on $r = 5$, the two-sided conditions $p_i \pm 1 \mid n \pm 1$, the conjugation $(\zeta_5 - 1)^{p^2} = -\zeta_5^{-1}(\zeta_5 - 1)$ and the bound $T_p \mid 10(p^2 - 1)$ are all in the Lenstra–Pomerance notes [10]. The paradigm of studying subgroups of $(\mathbb{F}_p[X]/(\Phi_r))^\times$ generated by cyclotomic elements such as $X - 1$, with lower bounds on their orders, is Popovych's [12, 13]; our set $S(p, r)$ is a specific formulation within that paradigm, not a new idea. The reduction of the congruence to a CRT system of Korselt-type conditions appears in Váňa's 2009 student paper [11], which deserves to be better known; our Sections 6–7 sharpen it to the field component Φ_5, prove the *necessity* of the explicit system on the minimal skeleton (via the rigidity theorem), and determine the exact order decomposition. The quintic character of the golden ratio is classical (Williams–Hardy [18], building on E. Lehmer [16] and Williams [17]); the translation between power residuosity and splitting in Kummer extensions is completely classical (see e.g. Helou [19] for norm-residue symbols and cyclotomic units), as is the Fibonacci torus of Cubre–Rouse [20]. The support theorem of Corrales–Rodrigáñez–Schoof [21] has different quantifiers from the moving-exponent finite-support problem here. Primitive-divisor theorems for Lucas and Lehmer sequences [23, 24], and Breuer's work on orders of Gauss periods [22], are likewise close in method but do not directly impose the class-specific restrictions required below.

What we have not found in the literature examined is: the moment theory applied to this congruence (Section 3); the identification $M_2 = 3\,\mathrm{ind}_5(\varepsilon)$ of the Agrawal moment with the quintic character of the golden unit; the necessary-and-sufficient two-lock classification of the local exceptions at $r = 5$; the mod-5 structure theorem for counterexamples; and the derivation of the $(+1)$-side Lenstra–Pomerance conditions, with their exact weakening measured by a Kummer defect, from the local algebra. These bridges, not the individual ingredients, are the contribution claimed here; we would welcome pointers to earlier occurrences.

Disclosure and verification. The results were found and assembled in the UNICO/NOUS multi-model pipeline, initiated and operated by Daniele Cappello, who curated the public artifacts and accepts responsibility for their provenance, stated scope, and release. This process responsibility is not a claim of specialist mathematical verification. Model-based adversarial review was used; no human specialist review is claimed. Results labelled as theorems are mathematical claims of the paper; only the subset named in the repository's claim-status table is additionally checked end to end by Lean. Several deeper results are presented with proof architectures whose ordinary mathematical review remains independent of the kernel layer. The accompanying Lean 4 implementation verifies a stated elementary kernel layer over Mathlib with kernel-pure proofs (no `sorry` in the submitted implementation and no extra axioms). Separate Mathlib-only challenge statements contain explicit proof holes and are checked against their solution files by a pinned version of `leanprover/comparator`; those trusted statements are not counted as implementation proofs. The Lean layer does not silently stand for the analytic or finite-field arguments left on paper. Computational statements always carry their exact domain; they are falsification controls, not proofs, except where stated as certified exclusions in explicitly bounded regions.

For the four-coefficient reduction, the kernel now defines the coefficients by polynomial remainder, checks simultaneous vanishing in both split components, proves the canonical-signature uniqueness, and derives the exact scalar orders from $p \mid D_{r,s}$. The final identification with the integral integer B_s remains paper-level, and the class-specific implication (4.8) remains precisely H4; neither is represented by a hidden axiom.

For the golden moment, the kernel layer now also closes the previously separate product-to-sum interface. The declaration `quintic_index_indexedMomentUnit_eq_moment` proves that applying a quintic index to the weighted multiplicative unit gives the additive moment M_j; `indexedMomentUnit_two_eq_quadraticMomentUnit` enumerates $(\mathbf{Z}/5\mathbf{Z})^\times$ and proves the weights $1, 4, 4, 1$; and `cyclotomic_quadratic_moment_eq_three_golden_index` composes this with the golden factorization. The declaration `localS5_unit_rows` starts from the literal quotient-ring predicate `LocalS5`, evaluates the four primitive fifth-root rows, and `localS5_canonical_golden_moment_obstruction` constructs the cyclotomic units internally and applies one and the same quintic character to obtain covariance and the golden obstruction. Finally, `localS5_canonical_quintic_locks_of_ne_one` formalizes the direct conclusion that a nonidentity residue coordinate annihilates both the scalar-five and golden-unit indices for that same character. These declarations use no published index table, discrete-logarithm generator, or computer-algebra transcript as a premise. The analogous derivation from the complete external definition of $S(p, r)$ for general r remains a separate paper-level interface.

2. The local set and its discrete-logarithm form

Fix distinct primes p, r with r odd. Let $k = \mathrm{ord}_r(p)$, $q = p^k$, $N = q - 1$, $K = \mathbb{F}_q$, and let $\zeta \in K^\times$ have order r. Set $G = (\mathbb{Z}/r)^\times$, $h = r - 1$, and for $a \in G$ put $u_a = \zeta^a - 1 \in K^\times$, $e_a = \log_\gamma(u_a) \in \mathbb{Z}/N$ for a fixed generator γ of $K^\times$. Let

$$S = S(p, r) = \{\, m \in \mathbb{Z}^+ : (X - 1)^m \equiv X^m - 1 \text{ in } A = \mathbb{F}_p[X]/(\Phi_r) \,\}.$$

Lemma 2.1 (coprimality)**.** *$m \in S \implies \gcd(m, r) = 1$.*

Theorem 2.2 (discrete-logarithm form)**.** *$m \in S$ iff $\gcd(m, r) = 1$ and $m\, e_a \equiv e_{am \bmod r}$ $\pmod N$ for every $a \in G$.*

Sketch. $A \cong \prod K$ over Frobenius orbits of primitive r-th roots; the congruence in each component reads $(\zeta^a - 1)^m = \zeta^{am} - 1$, which in $\log_\gamma$-coordinates is the displayed congruence; Frobenius transports the identity across an orbit. The complete general-r decomposition and discrete-logarithm equivalence are proved here at paper level. For $r = 5$, the project repository separately gives a kernel-checked specialization: `localS5_unit_rows` derives the four rows from the literal quotient-ring predicate `LocalS5`, and `localS5_quintic_covariance` applies a quintic character directly, without choosing a discrete-logarithm generator. □

Lemma 2.3 (squarefree reduction)**.** *Let $n \geq 2$ be squarefree with $r \nmid n$. Then $(X - 1)^n \equiv X^n - 1 \pmod{n, X^r - 1}$ iff $n/p \in S(p, r)$ for every prime $p \mid n$.*

Sketch. CRT over $p \mid n$; injectivity of the absolute Frobenius on $\mathbb{F}_p[X]/(X^r - 1)$ replaces $(X - 1)^n$ by $(X - 1)^{n/p}$; the component at $X \mapsto 1$ is trivial. For $r = 5$, the complete equivalence is kernel-checked as `squarefree_ingress_iff`. Its forward direction uses a fully explicit inverse-Frobenius evaluation: choose $ap \equiv 1 \pmod 5$ and evaluate at ζ_5^a, avoiding any abstract reducedness assumption on the quotient ring. □

We also record that $S \bmod M$ is a subgroup of $(\mathbb{Z}/M)^\times$ for $M = \mathrm{lcm}(T, r)$, T the multiplicative order of $X - 1$ in $A^\times$; membership in S depends exactly on $m \bmod M$ (periodicity), and $p \in S$ always (Frobenius).

3. Cyclotomic moments

Since $r \mid N$, the reductions $\bar{e}_a = e_a \bmod r \in \mathbb{F}_r$ are well defined. For $0 \leq j \leq h - 1$ define

$$M_j = \sum_{a \in G} a^j\, \bar{e}_a \in \mathbb{F}_r, \qquad J = \{j : M_j \neq 0\}, \qquad T = \{t \in \mathbb{F}_r^\times : t^{j+1} = 1\ \forall j \in J\}.$$

Theorem 3.1 (moment obstruction)**.** *$\mathrm{im}_r\, S \subseteq T$. Moreover*

$$T = \mu_e, \qquad e = \gcd\big(h, \gcd_{j \in J}(j + 1)\big).$$

Proof. For $m \in S$ with $t = m \bmod r$: reducing Theorem 2.2 mod r gives $t\, \bar{e}_a = \bar{e}_{ta}$. Multiply by a^j, sum over a, and substitute $b = ta$: $tM_j = t^{-j} M_j$, so $(t - t^{-j}) M_j = 0$; if $M_j \neq 0$ then $t^{j+1} = 1$. □

Theorem 3.2 (structure of the moments)**.** *(i) $M_j \neq 0 \implies p^{j+1} \equiv 1 \pmod r$. (ii) $M_j = 0$ for odd $j \neq h - 1$. (iii) $M_{h-1} = -d\bar{w}/2$ with $d = (N/r) \bmod r$; in particular $M_{h-1} = 0$ iff $r^2 \mid N$, and its associated constraint is vacuous. (iv) $M_0 = \log_\gamma(r) \bmod r$; thus $M_0 = 0$ iff $r \in (K^\times)^r$.*

The choices of γ and ζ only scale the moments by units, so J, T and all vanishing conditions are invariant. Part (iv) uses $\prod_a(\zeta^a - 1) = \Phi_r(1) = r$. For $r = 3$, composing (iv) with the classical cubic-residuacity criterion ($4p = L^2 + 27M^2$; 3 is a cubic residue iff $3 \mid M$, Jacobi; see [15]) characterizes the degenerate primes exactly.

Computational verification 3.3. Zero violations of Theorems 3.1–3.2 over all (p, r) with $r \in \{3, 5, 7, 11, 13\}$, $q = p^k \le 5 \cdot 10^5$, and in the $k = 1$ ranges $p < 10^6$ ($r \le 7$), $p < 2 \cdot 10^5$ ($r \le 19$). All 406 historically known local exceptions (397 at r=3, 8 at r=5, 1 at r=7) have all even moments vanishing, as the theory predicts.

4. $r = 5$: the golden unit

From now on $r = 5$, $p \equiv 1 \pmod 5$ in this section. Fix z of order 5 in $\mathbb{F}_p^\times$, coordinate $\sqrt{5} = 1 + 2(z + z^{-1})$, and set $\varepsilon = (1 + \sqrt{5})/2$, the golden unit. Fix a quintic character $\chi : \mathbb{F}_p^\times \to \mathbb{F}_5$ (equivalently, fix one discrete-logarithm normalization modulo 5), and write $\mathrm{ind}_5(x) = \chi(x)$. The same character is used in the definitions of M_2 and $\mathrm{ind}_5(\varepsilon)$. Changing the normalization multiplies both sides of the identity below by the same nonzero scalar, so its vanishing content is canonical. In the Williams–Hardy certificate this choice is coordinated by a primitive generator g and $z = g^{(p-1)/5}$.

Theorem 4.1 (golden theorem)**.** $M_2 = 3 \cdot \mathrm{ind}_5(\varepsilon)$. *In particular* $M_2 = 0 \iff \varepsilon \in (\mathbb{F}_p^\times)^5$.

Proof. $M_2 = \mathrm{ind}_5(U_2)$ with $U_2 = \prod_{a=1}^4 (z^a - 1)^{[a^2 \bmod 5]} = AB^4$, where $A = (z - 1)(z^4 - 1) = (5 - \sqrt{5})/2$ and $B = (z^2 - 1)(z^3 - 1) = (5 + \sqrt{5})/2$. Then $AB = 5$, so $U_2 = 5B^3 = (\sqrt{5})^5 \varepsilon^3$ since $B = \sqrt{5}\,\varepsilon$; the factor $(\sqrt{5})^5$ is a fifth power. $\square$

The other choice of $\sqrt{5}$ maps $\varepsilon \mapsto -\varepsilon^{-1}$ and does not affect the vanishing (-1 is a fifth power). Combined with Theorem 3.2(iv), a nontrivial local image at $r = 5$ requires 5 *and* ε to be simultaneous quintic residues. Via the Dickson system of quintic cyclotomy, $\mathrm{ind}_5(\varepsilon) \equiv 3v - u$ and $\mathrm{ind}_5(5) \equiv -2u - v \pmod 5$; the criterion for ε is classical (Williams–Hardy [18], Thm 5 and Remark 2), while its identification with M_2 is the candidate new bridge claimed here: it was not found in the targeted search, and priority remains provisional pending specialist review. As a normalization check independent of the Lean proof, a tracked replay computes M_2 directly for every one of the 22 rows in Williams–Hardy Table 5 and obtains

$$M_2 = 3\ \mathrm{ind}_5(\varepsilon) \pmod 5$$

in 22/22 cases. This finite comparison checks conventions, signs and the factor 3; it is neither a proof premise nor evidence of historical priority. A second independent replay enumerates all 1,158,464 relevant exponent classes for the 306 primes $p \equiv 1 \pmod 5$, $p < 10^4$, and finds zero violations of the local-row, same-character covariance and golden-obstruction chain. This is again a finite regression, not a universal proof. The pair (M_0, M_2) is a linear coordinate system on the Dickson plane (u, v) mod 5 (determinant 4), so the degenerate locus $M_0 = M_2 = 0$ is exactly $u \equiv v \equiv 0$.

Theorem 4.2 (lifted moments and the tower)**.** *Let* $s \ge 1$ *with* $5^s \mid p - 1$ *and define* $M_j^{(s)}$ *with Teichmüller weights. If* $m \in S$ *has* $t = m \bmod 5 \neq 1$, *then* $M_0^{(s)} \equiv M_2^{(s)} \equiv 0 \pmod{5^s}$. *Moreover* $M_0^{(s)} = \mathrm{ind}_{5^s}(5)$ *and* $M_2^{(s)} = -2\ \mathrm{ind}_{5^s}(\varepsilon)$. *Hence* $t \neq 1$ *forces* $5, \varepsilon \in (\mathbb{F}_p^\times)^{5^s}$ *for every* $s \le v_5(p - 1)$.

Theorem 4.3 (golden Kummer tower)**.** *Let* $K_s = \mathbb{Q}(\zeta_{5^s})$, $L_s = K_s(5^{1/5^s}, \varepsilon^{1/5^s})$, $B_s = K_s(5^{1/5^{s-1}}, \varepsilon^{1/5^{s-1}})$. *For every* $s \ge 1$: $[L_s : K_s] = 5^{2s}$ *and* $[L_s : B_s] = 25$.

Sketch. $5 \notin K_s^{\times 5}$: otherwise K_s would contain the splitting field of $X^5 - 5$, which is non-abelian over $\mathbb{Q}$, contradicting $K_s/\mathbb{Q}$ abelian. The same argument applies to ε through $X^{10} - X^5 - 1 = (X^5 - \varepsilon)(X^5 - \varepsilon')$ with $\varepsilon' = -\varepsilon^{-1}$, whose splitting field is again non-abelian. Exact order 5^s of the classes and Kummer independence of 5 and ε (using the conjugation $\sigma(\varepsilon) = -\varepsilon^{-1}$) complete the proof. □

By Kummer–Dedekind splitting and Chebotarev, among the primes that have passed floor $s-1$, those passing floor s have conditional density $1/25$: each floor of the tower removes $24/25$ of the survivors. This explains quantitatively the observed rarity of local exceptions (e.g. the factor $\sim$20 drop of the degeneracy rate in the branch $p \equiv 1 \pmod{25}$).

5. The two locks and the local classification

Still $r = 5$, $p \equiv 1 \pmod 5$, $N = p - 1$, $s = v_5(N)$, $v = v_2(N)$. The system for $t = -1$ has an obstruction only at $\ell = 2$ and $\ell = 5$; all other ℓ-adic components are automatically solvable.

Lemma 5.1 (binary lock)**.** *Set $\nu_a = \min(v, v_2(e_a))$. The four rows at $t = -1$ are compatible iff $\nu_1 = \nu_2 = \nu_3 = \nu_4 < v$; by the conjugate-pair relations $e_4 \equiv e_1 + 2^{v-1}(\mathrm{odd})$, $e_3 \equiv e_2 + 2^{v-1}(\mathrm{odd}) \pmod{2^v}$, this reduces to $\nu_1 = \nu_2 \le v - 2$.*

Corollary 5.2. *$v_2(p-1) \ge 2$ is necessary for a local exception; with $p \equiv 1 \pmod 5$ this gives $p \equiv 1 \pmod{20}$.*

Theorem 5.3 (local classification at $r = 5$)**.** *$\mathrm{im}_5 S$ is a subgroup of $\mathbb{F}_5^\times$; any nontrivial subgroup contains -1, so $\mathrm{im}_5 S \neq \{1\}$ iff $-1 \in \mathrm{im}_5 S$, and*

$$\mathrm{im}_5 S \neq \{1\} \iff \Big[5,\ \varepsilon \in (\mathbb{F}_p^\times)^{5^s}\Big] \ \wedge\ \Big[\nu_1 = \nu_2 \le v_2(p-1) - 2\Big].$$

For residues of order 4 the situation is drastically stiffer: for $t \in \{2, 3\}$ *every* odd prime divisor $\ell \mid p-1$ imposes a lock (equal truncated valuations and a square condition on the unit parts), so heuristically the probability decays like $\prod_{\ell \mid p-1} O(1/\ell)$. In a census of all 71 local exceptions with $p < 10^6$, none has an order-4 element: $\mathrm{im}_5 S \in \{\{1\}, \{1, -1\}\}$ throughout the censused domain. Two further exact results constrain order 4: in the quadratic branch ($p \equiv -1 \bmod 5$) existence of $t = 2$ is *decidable* by the criterion $g_1 = g_2 \wedge c^2 \equiv p \pmod T$ (censused: 0/104); in the split branch the "golden compression" forces $5^{m-1} = 1$, $\varepsilon^{m+1} = -1$, whose CRT solvability profile ($4 \mid E$, $5 \nmid RE$, $\gcd(R, E) \mid E/2 - 2$, with $R = \mathrm{ord}_p 5$, $E = \mathrm{ord}_p \varepsilon$) is realized by *no* prime $p < 10^7$ (0 of 166,104). These motivate:

Hypothesis 5.4 (H4)**.** For every prime $p \equiv \pm 1 \pmod 5$, $\mathrm{im}_5 S(p, 5)$ contains no element of order 4.

5.1. A prime-support formulation of H4.

For $n \equiv 4 \pmod 5$, define

$$A_n = \begin{cases} L_n, & n \text{ even}, \\ F_n, & n \text{ odd}, \end{cases} \qquad H_n = \gcd(A_n, 5^{n-1} + 1), \tag{4.1}$$

where F_n, L_n are the Fibonacci and Lucas sequences.

Theorem 5.5 (scalar completeness and the sequence H_n)**.** *Let $p \neq 2, 5$ be split in $\mathbb{Q}(\sqrt{5})$. The local image $\mathrm{im}_5 S(p, 5)$ contains an element of order 4 if and only if there is an $n \equiv 4 \pmod 5$ such that*

$$5^{n-1} \equiv -1 \pmod p, \qquad \varepsilon^{2n} \equiv -1 \pmod p. \tag{4.2}$$

Equivalently, $p \mid H_n$ *for some* $n \equiv 4 \pmod 5$*. Consequently H4 is equivalent to the assertion that every prime in*

$$\bigcup_{n\equiv 4\ (5)} \mathrm{Supp}_{(10)}(H_n) \tag{4.3}$$

is inert in $\mathbb{Q}(\sqrt{5})$*.*

The point of "scalar completeness" is that no independent cyclotomic direction remains hidden in (4.2). In the linear branch the four defects u_a^{2n-1}/u_{2a} coincide; their common sign can be repaired by shifting the exponent by half the order of the cyclotomic subgroup. In the quadratic branch the same two scalar equations force the intrinsic conditions $\langle u_1\rangle = \langle u_2\rangle$ and $\theta^2 = \mathrm{Frob}_p$. Thus (4.2) is equivalent to the complete local row, not only necessary for it.

Theorem 5.6 (support of H_n)**.** *Every prime* $p \neq 2, 5$ *inert in* $\mathbb{Q}(\sqrt{5})$ *divides* H_n *for infinitely many* $n \equiv 4 \pmod 5$*. If* $p \equiv 2 \pmod 5$*, then the canonical index is*

$$p \mid H_{(p+1)/2}. \tag{4.4}$$

Hence H4 is equivalent to equality in (4.3) between the support of the H_n *and the set of inert primes outside* $\{2, 5\}$*.*

Proof of the support assertion. For an inert p, Frobenius sends ε to $-\varepsilon^{-1}$, so $\varepsilon^{p+1} = -1$, while Euler's criterion gives $5^{(p-1)/2} = -1$. Thus $n_0 = (p+1)/2$ satisfies both scalar equations. All simultaneous solutions form one progression modulo

$$\mathrm{lcm}(\mathrm{ord}(\varepsilon)/2, \mathrm{ord}_p(5))\,.$$

This modulus is coprime to 5, so the progression meets $n \equiv 4 \pmod 5$ infinitely often. When $p \equiv 2 \pmod 5$, n_0 already has the required residue. $\square$

A certified census of all 20,000 indices $n \leq 100{,}000$ with $n \equiv 4 \pmod 5$ found 9,725 nontrivial H_n and 4,522 distinct prime factors, all inert. Of these, 4,503 are the canonical factors $p \equiv 2 \pmod 5$ forced by (4.4), and 19 are translated inert factors. Thus the informative negative observation is zero split factors, not the raw number of factors. A separate prime-first census found no H-profile below 10^9.

For comparison, the positive-sign companion J_n can have split odd factors. Three exact J-profiles occur below 10^9, each saturating $\mathrm{ord}_p(5)\,\mathrm{ord}_p(\varepsilon) = (p-1)/2$. The earlier proposed closed formula for J_n is false; the valid theorem is only that no inert odd prime divides J_n. A kernel-checked proof is elementary. If the index is even, the positive scalar equation writes 5 as an explicit square; if it is odd, the inert semiperiod $\varepsilon^{p+1} = -1$ and the positive equation force $1 = -1$. This correction does not affect Theorems 5.5 and 5.6.

5.2. **A four-coefficient primitive support.** The norm of an algebraic integer detects only one of the two prime ideals above a split rational prime. The following refinement removes that ambiguity. For an admissible pair

$$\gcd(r, s) = 1, \qquad r + s \equiv 1 \pmod 2, \qquad 5 \nmid rs,$$

put $m = 4rs$. The simultaneous congruences

$$k \equiv 1 \pmod{2r}, \qquad k \equiv -1 \pmod{2s}$$

have two lifts modulo m. Exactly one makes both complementary quotients odd: if r is even, the two lifts change $(k-1)/(2r)$ by the odd number s; if s is even, they change $(k+1)/(2s)$ by the odd number r. Coprimality of r, s then shows that the two quotients are already coprime to s and r, respectively (subtract the equations defining the two congruences). Thus this lift is the unique $k = k_{r,s} \pmod m$ with

$$\gcd(m, k-1) = 2r, \qquad \gcd(m, k+1) = 2s. \tag{4.5}$$

Set

$$\gamma = \frac{5+\sqrt{5}}{2}, \qquad \gamma' = 5 - \gamma,$$

and define $U_0 = 0, U_1 = 1, U_{j+2} = 5U_{j+1} - 5U_j$, so that

$$\gamma^j = U_j\gamma - 5U_{j-1}.$$

Writing

$$\Phi_m(\gamma) = c_{r,s} + d_{r,s}\gamma, \qquad A_{r,s} = U_k + 1, \qquad W_{r,s} = U_{k-1} + 1,$$

and, with $x = \varepsilon^2$, set

$$C_r = \Phi_{2r}(5), \qquad B_s = x^{-\varphi(2s)/2}\Phi_{2s}(x) \in \mathbb{Z}, \qquad G_{r,s} = \gcd(C_r, B_s),$$

where $B_1 = 1$. Finally define

$$D_{r,s} = \gcd(c_{r,s}, d_{r,s}, A_{r,s}, W_{r,s}). \tag{4.6}$$

Lemma 5.7 (exact-level integers)**.** *Let $p \nmid 10rs$ be prime. Then*

$$p \mid C_r \iff \operatorname{ord}_p(5) = 2r, \qquad p \mid B_s \iff \operatorname{ord}_p(\varepsilon^2) = 2s.$$

For $s > 1$, the displayed definition of B_s is an ordinary rational integer; for $s = 1$, both sides of the second equivalence are false.

Proof. The assertion for $C_r = \Phi_{2r}(5)$ is the standard exact-level criterion for cyclotomic polynomials, since $p \nmid 2r$.

Put $x = \varepsilon^2$. It is an algebraic unit, its conjugate is x^{-1}, and, for $s > 1$, $\varphi(2s)$ is even. Reciprocity of the cyclotomic polynomial gives

$$\Phi_{2s}(X^{-1}) = X^{-\varphi(2s)}\Phi_{2s}(X).$$

Hence $x^{-\varphi(2s)/2}\Phi_{2s}(x)$ is fixed by conjugation. It is also an algebraic integer, because x is a unit, and therefore belongs to $\mathbb{Q} \cap \mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathbb{Z}$. After reduction modulo any prime above p, the normalizing power of x is invertible. Thus $p \mid B_s$ is equivalent to $\Phi_{2s}(x) = 0$. Since $p \nmid 2s$, the factorization of $X^{2s} - 1$ is separable and this is equivalent to x having exact order $2s$. The two split components are conjugate and therefore have the same order. Finally, if $s = 1$, order 2 would give $x = -1$; the relation $x + x^{-1} = 3$ would then force $p = 5$, which is excluded, while $B_1 = 1$. □

Theorem 5.8 (four-coefficient primitive support)**.** *For every prime $p \nmid 10rs$,*

$$p \mid G_{r,s} \iff p \mid D_{r,s}, \tag{4.7}$$

where $G_{r,s}$ is the primitive cyclotomic gcd associated with the orders $2r$ of 5 and $2s$ of ε^2. Consequently H4 is equivalent to

$$p \mid D_{r,s}, \quad p \nmid 10rs \implies \left(\frac{5}{p}\right) = -1. \tag{4.8}$$

Proof. Lemma 5.7 gives $\operatorname{ord}(5) = 2r, \operatorname{ord}(\varepsilon^2) = 2s$. In the cyclic subgroup generated by $\sqrt{5}$ and ε, both $\gamma = \sqrt{5}\,\varepsilon$ and $\gamma' = \sqrt{5}\,\varepsilon^{-1}$ have exact order m, and (4.5) gives

$$\gamma' = \gamma^k, \qquad k^2 \equiv 1 \pmod{m}.$$

Thus $\Phi_m(\gamma)$ and $\gamma^k - \gamma'$ vanish. In the inert case, linear independence of $1, \gamma$ gives divisibility of all four coefficients. In the split case the second component is essential: conjugation and $k^2 \equiv 1 \pmod{m}$ give $(\gamma')^k = \gamma$, so both algebraic integers vanish in both components of $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}/p \simeq \mathbb{F}_p \times \mathbb{F}_p$. Since $p \neq 5$, the integral basis is nondegenerate and again $p \mid c, d, A, W$.

Conversely, divisibility of the four coefficients gives $\Phi_m(\gamma) = 0$ and $\gamma^k = \gamma'$ in every component. Since $p \nmid m$, γ has exact order m, and hence

$$\operatorname{ord}(5) = \frac{m}{\gcd(m, k+1)} = 2r, \qquad \operatorname{ord}(\varepsilon^2) = \frac{m}{\gcd(m, k-1)} = 2s.$$

This recovers $p \mid G_{r,s}$. □

Theorem 5.9 (level reciprocity)**.** *Let $p \nmid 10rs$ be a split prime divisor of $D_{r,s}$, and write*

$$p - 1 = 4rs\,h.$$

Then

$$\boxed{h \equiv 0 \pmod{2} \quad\Longleftrightarrow\quad p \equiv 1 \pmod{5} \quad\Longleftrightarrow\quad 5 \mid h.} \tag{4.9}$$

Consequently $10 \mid h$ in the $p \equiv 1 \pmod{5}$ branch, whereas $\gcd(h, 10) = 1$ in the $p \equiv 4 \pmod{5}$ branch. Equivalently, only the five residue classes

$$h \equiv 0, 1, 3, 7, 9 \pmod{10}$$

remain possible globally. For a fixed pair (r, s), the restriction is sharper: the branch $p \equiv 1 \pmod{5}$ has $h \equiv 0 \pmod{10}$, while in the branch $p \equiv 4 \pmod{5}$ one has

$$\begin{array}{c|cccc} rs \bmod 5 & 1 & 2 & 3 & 4 \\ \hline h \bmod 10 & 7 & 1 & 9 & 3. \end{array} \tag{4.10}$$

In particular, the corresponding elementary lower bounds for p are

$$p \geq 28rs + 1, \quad 4rs + 1, \quad 36rs + 1, \quad 12rs + 1,$$

respectively.

Proof. Let $\gamma \in \mathbb{F}_p$ be either root of $\gamma^2 - 5\gamma + 5 = 0$. A direct Gaussian-period argument gives

$$-\gamma \in (\mathbb{F}_p^\times)^2 \quad\Longleftrightarrow\quad p \equiv 1 \pmod{5}. \tag{4.11}$$

Indeed, over a field containing a fifth root of unity ζ, the two roots are $-(\zeta - \zeta^{-1})^2$ and $-(\zeta^2 - \zeta^{-2})^2$. Conversely, from a square root v of $-\gamma$, the element

$$z = \frac{\gamma - 3 + v}{2}$$

satisfies $z^5 = 1$ and $z \neq 1$; hence $5 \mid p - 1$.

Theorem 5.8 gives $\operatorname{ord}_p(\gamma) = m = 4rs$. Since $\gamma^{m/2} = -1$ and $m/2$ is even, Euler's criterion yields

$$(-\gamma)^{(p-1)/2} = (-1)^h.$$

Thus $-\gamma$ is a square exactly when h is even. Finally $5 \nmid m$, so $5 \mid p - 1 = mh$ exactly when $5 \mid h$. Combining these three equivalences proves (4.9). The assertions modulo 10 follow immediately. In the $p \equiv 4 \pmod{5}$ branch,

$$4rs\,h \equiv p - 1 \equiv 3 \pmod{5}.$$

Combining this congruence with the oddness of h gives the four entries of (4.10), and the lower bounds follow from $p = 4rs\,h + 1$. □

Corollary 5.10 (order-product barrier)**.** *Under the hypotheses of Theorem 5.9, in the branch* $p \equiv 1 \pmod 5$,

$$\boxed{10\ \mathrm{ord}_p(5)\ \mathrm{ord}_p(\varepsilon^2) \mid p-1.} \tag{4.12}$$

Proof. Theorem 5.8 gives $\mathrm{ord}_p(5) = 2r$ and $\mathrm{ord}_p(\varepsilon^2) = 2s$, so their product is $4rs$. By Theorem 5.9, $p - 1 = 4rs\,h$ with $10 \mid h$ in this branch. □

Theorem 5.11 (residual power depth)**.** *Under the hypotheses of Theorem 5.9, put* $m = 4rs$, *and let* $d > 0$ *divide* m. *Then*

$$\boxed{-\gamma \in (\mathbb{F}_p^\times)^d \iff d \mid h.}$$

If d is even, then equivalently

$$\boxed{-\gamma \in (\mathbb{F}_p^\times)^d \iff \mathrm{lcm}(d, 5) \mid h.}$$

In particular,

$$-\gamma \in (\mathbb{F}_p^\times)^4 \iff 20 \mid h. \tag{4.13}$$

Proof. The element γ has exact order m in the cyclic group $\mathbb{F}_p^\times$, whose order is mh. Since $4 \mid m$, $\gamma^{m/2} = -1$, and

$$-\gamma = \gamma^{m/2+1}.$$

Moreover $\gcd(m, m/2 + 1) = 1$, so $-\gamma$ also has exact order m. For $d \mid m$, the standard power-map criterion in a finite cyclic group gives

$$-\gamma \in (\mathbb{F}_p^\times)^d \iff (-\gamma)^{mh/d} = 1 \iff m \mid \frac{mh}{d} \iff d \mid h.$$

If d is even, then $d \mid h$ forces h even; by Theorem 5.9, this is equivalent in the good range to $5 \mid h$. Thus $d \mid h$ is equivalent to $\mathrm{lcm}(d, 5) \mid h$. The quartic case is $d = 4$. □

The intersection ideal

$$\mathfrak{I}_{r,s} = \big(\Phi_m(\gamma), \gamma^k - \gamma'\big) \subset \mathcal{O}_{\mathbb{Q}(\sqrt{5})}$$

is generated, in the basis $1, \gamma$, by the four columns

$$(c, d), \quad (-5d, c + 5d), \quad (-5W, A), \quad (-5A, 5(A - W)).$$

Its index is therefore the gcd of the six 2×2 minors. Away from $10rs$, its radical is the radical of $D_{r,s}\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$. No universal equality of multiplicities is claimed.

Theorem 5.12 (the first four single odd-prime support depths are impossible)**.** *No split H-profile prime satisfies*

$$p - 1 = 2^b q^e, \qquad b \in \{3, 4, 5, 6\},$$

with q an odd prime and $e \geq 1$.

Proof. Write $\mathrm{ord}_p(5) = 2r$, $\mathrm{ord}_p(\varepsilon^2) = 2s$, with $\gcd(r, s) = 1$. Since the two semiorders divide $2^{b-1}q^e$, the prime q cannot divide both, and the semiorder on at least one side is a power of two. Thus one of the two complete orders divides 2^b.

For $b = 3$, if the base-5 order is 2-primary, then

$$p \mid 5^8 - 1 = 2^5 \cdot 3 \cdot 13 \cdot 313,$$

whose good prime factors are all inert. If the golden order is 2-primary, then

$$p \mid \left| N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}\big((\varepsilon^2)^8 - 1\big) \right| = 2205 = 3^2 \cdot 5 \cdot 7^2,$$

again with no good split factor.

The next three depths are finite exact resultant calculations. The following table records every split factor not already eliminated by inertia or by $p \in \{2, 5\}$:

2^b	base-5 side	golden side
16	$11489, \quad (11489-1)/16 = 718$	$\varnothing$
32	$11489, \quad (11489-1)/32 = 359$	$\varnothing$
64	$641,\ 11489,\ 29423041,\ 241931001601$	4481

Here the factorizations used by the kernel are

$$\begin{aligned}
5^{16} - 1 &= 2^6 \cdot 3 \cdot 13 \cdot 17 \cdot 313 \cdot 11489, \\
5^{32} - 1 &= 2^7 \cdot 3 \cdot 13 \cdot 17 \cdot 313 \cdot 2593 \cdot 11489 \cdot 29423041, \\
5^{32} + 1 &= 2 \cdot 641 \cdot 75068993 \cdot 241931001601, \\
\left| N((\varepsilon^2)^{32} - 1) \right| &= 3^2 \cdot 5 \cdot 7^2 \cdot 47^2 \cdot 2207^2, \\
\left| N((\varepsilon^2)^{32} + 1) \right| &= 1087^2 \cdot 4481^2.
\end{aligned}$$

At depth 16, the displayed cofactor 718 is even. At depth 32, the only arithmetically compatible split factor is 11489. There $\mathrm{ord}_{11489}(5) = 16$, hence $r = 8$ and $q^e = 359$. Coprimality forces $s \in \{1, 359\}$. The first choice would make ε^2 have order 2, while the second would give $(\varepsilon^2)^{718} = 1$; both are impossible, the latter because the exact kernel certificate gives

$$(\varepsilon^2)^{718} = -1 \pmod{11489}$$

for either root of $X^2 - 3X + 1$.

At depth 64, the cofactors attached to 641, 241931001601, and 4481 are even, $11489 - 1$ is not divisible by 64, and

$$\frac{29423041 - 1}{64} = 459735 = 3 \cdot 5 \cdot 30649$$

is not a prime power. These exhaust the split factors in the exact factorizations above, completing all four depths. □

Theorem 5.13 (5-smooth split support is impossible)**.** *No split H-profile prime in the branch* $p \equiv 1 \pmod 5$ *satisfies*

$$p - 1 = 2^b 5^f$$

for any $b, f \geq 0$.

Proof. Write the exact scalar orders as $\mathrm{ord}_p(5) = 2r$ and $\mathrm{ord}_p(\varepsilon^2) = 2s$. The primitive level $4rs$ is prime to 5, while both orders divide $p - 1$. Hence $r, s \mid 2^b$. Since $r + s$ is odd, one of r, s is odd, and an odd divisor of a power of two is 1.

If $r = 1$, then 5 has order 2, so $5 \equiv -1 \pmod p$; thus $p \mid 6$, which is incompatible with $p \equiv 1 \pmod 5$. If $s = 1$, then $x = \varepsilon^2$ has order 2, so $x = -1$. But $x^2 - 3x + 1 = 0$, and substitution gives $5 = 0$, contrary to $p \neq 5$. □

An independent exact replay on the 244 admissible pairs $r, s \leq 30$ found no discrepancy in (4.7) or in the localized ideal radical. A separate smooth-$p-1$ falsifier enumerated all

$$p - 1 = 2^b 5^f m \leq 10^{15}$$

with m supported on $\{3, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43\}$: among 4,253,287 prime values and 161,918 survivors of the dyadic lock, it found no complete H-profile. This is a finite test of a structured family, not a proof of H4.

A complete prime-first scan of $10^9 \le p < 10^{10}$ examined 202,100,126 split primes and exactly factored all 9,622,566 survivors of the dyadic lock, again finding no complete H-profile. The minimum gcd of the two semiorders is 3, attained at

$$p = 1{,}368{,}322{,}369, \qquad \mathrm{ord}_p(5) = 6{,}108{,}582, \qquad \mathrm{ord}_p(\varepsilon^2) = 672.$$

Thus even this closest recorded case still fails the odd coprimality condition. The conditioned expected mass of the scanned interval is only 0.217780; its zero count is a finite falsification result, not evidence for a universal common divisor and not a proof of H4.

The candidates are, at least, provably rare:

Theorem 5.14 (density zero of H4 candidates)**.** *In each branch, the set of primes satisfying the mobile Kummer covering condition of the unified H4 profile (for every odd prime $\ell \mid p-1$, $\ell \neq 5$: 5 or ε is an ℓ-th power mod p) has relative natural density zero, hence absolute density zero. Proof idea: for the level-one events $B_\ell = \{\ell \mid p-1,\ 5 \notin (\mathbb{F}_p^\times)^\ell,\ \varepsilon \notin (\mathbb{F}_p^\times)^\ell\}$ one computes $\delta(B_\ell) = (\ell-1)/\ell^2$ by Chebotarev on the rank-two Kummer fields of $(5, \varepsilon)$ over $\mathbb{Q}(\sqrt{5})$; the fiber-product disjointness for this family (proved by the same commutator argument as Theorem 12.3) makes finite families independent; and $\prod_{\ell \le Y}(1 - (\ell-1)/\ell^2) \to 0$ since $\sum (\ell-1)/\ell^2 = \infty$. Density zero does not assert finiteness, nor emptiness, nor H4 itself.*

Theorem 5.15 (quantitative compression under GRH)**.** *Assume Dedekind–GRH for the explicit joint Kummer composita*

$$\mathcal{K}_{i,d} = C_i \prod_{\ell \mid d}^{\mathrm{comp}} \mathbb{Q}(\sqrt{5}, \mu_\ell, 5^{1/\ell}, \varepsilon^{1/\ell}),$$

with d squarefree and $(d, 10) = 1$. In either split branch, the number of primes up to X satisfying the complete mobile H-profile obeys

$$N_{H,i}(X) \ll \frac{X}{(\log X)^2}. \tag{4.14}$$

The proof is a Selberg upper-bound sieve. Fiber-product disjointness gives the exact density factor $g(d) = \prod_{\ell \mid d}(\ell-1)/\ell^2$. A class-sensitive Grenié–Molteni estimate retains this factor, and the exact identity $\delta_i g(d)[\mathcal{K}_{i,d} : \mathbb{Q}] = \varphi(d)^2$ gives a summable remainder for the explicit choice $D = X^{1/7}$. This is a theorem under the stated GRH family, not an unconditional estimate.

H4 remains open. Its candidates have unconditional density zero and the GRH compression (4.14), but neither statement implies emptiness. The finite-level Frobenius classes of the scalar and cyclotomic systems remain nonempty at every tested and proved level. The pure metaplectic and spinorial routes do not see more than the already recorded residue data, and the full 2-power reciprocity tower permits the H-profile. These negative orientation theorems delimit the missing ingredient without supplying it.

6. Structure of squarefree counterexamples

Let n be squarefree composite, $5 \nmid n$, satisfying the congruence at $r = 5$, and write $\nu = n \bmod 5$, $\pi_i = p_i \bmod 5$, $t_i = (n/p_i) \bmod 5$ for its prime factors p_i (Lemma 2.3: $n/p_i \in S(p_i, 5)$).

Theorem 6.1 (mod-5 structure)**.** *(a) $t_i = \nu\pi_i^{-1}$: the local residues are determined. (b) If $\nu \in \{1,4\}$ then $n^2 \equiv 1 \pmod 5$; a counterexample has $\nu \in \{2,3\}$. (c) If $\pi_i \equiv \pm 1$ then t_i has order 4. (d) (unconditional) If n has an even number of prime factors, all $\equiv \pm 2 \pmod 5$, then $n^2 \equiv 1 \pmod 5$; hence a counterexample with even k has a factor $\equiv \pm 1$ whose local residue has order 4. (e) Under H4: every counterexample has an odd number $k \geq 3$ of factors, all $\equiv \pm 2 \pmod 5$, and $k_- = \#\{p_i \equiv -\nu\}$ satisfies $k_- \equiv (k-1)/2 \pmod 2$.*

The local parity seam and the H4-to-skeleton part of Theorem 6.1 are kernel-checked. More precisely, `localS5_orderFour_odd` proves that a good local row in residue class 2 or 3 mod 5 is automatically odd, while `squarefree_counterexample_concrete_dichotomy` proves that every squarefree candidate either has a split order-four witness or has all prime factors in the inert classes, with an odd number (at least three) of factors and with every complementary `LocalS5` row present. The kernel theorem `squarefree_counterexample_order_dichotomy` now continues this chain through the exact 1-or-p^2 congruence modulo $\mathrm{lcm}(\mathrm{ord}(\zeta_5 - 1), 5)$ at every skeleton factor, with the branch determined by whether $p_i \equiv n \pmod 5$. For each inert final row, the kernel now also carries the corresponding pure or twisted branch to divisibility by q^4 of the explicit integer resultant used in Section 10. The proof avoids a separate number-field norm interface: irreducibility of Φ_5 mod q makes Φ_5 divide the fiber polynomial modulo q, so the integral remainder is coefficientwise divisible by q, and resultant homogeneity supplies all four factors. Reduction modulo 5, where $\Phi_5 = (X-1)^4$, shows that each resultant is 1 mod 5; the kernel therefore also proves nonvanishing and $q^4 \leq |\mathrm{Res}|$. A complex-root product argument is now kernel-checked as well and gives $|\mathrm{Res}| \leq 16 \cdot 5^A$. Finally, `explicitResultantRows_of_skeleton` instantiates the formerly abstract skeleton-to-fiber interface: at every skeleton factor it selects the correct pure or twisted terminal resultant and supplies both bounds. Compatible-prefix CRT sufficiency and universal fiber emptiness remain paper-level/open. The exact conditional closure under exclusion of both walls is recorded as `no_squarefree_counterexample_of_no_split_and_no_explicit_rows`; its two open premises are visible in the theorem statement, and the two exclusion premises are quantified only over squarefree candidates.

In particular, under H4 the two-factor case of Conjecture 1.1 at $r = 5$ is settled: every squarefree n with at most two prime factors passing the test is prime or $n^2 \equiv 1 \pmod 5$. The reduction of a two-prime candidate to a split order-four witness is kernel-checked, without H4, as `two_prime_candidate_has_splitOrderFourWitness`; H4 is used only to exclude that final witness, in the kernel theorem `no_two_prime_candidate_of_localH4`.

Theorem 6.2 (rigidity in the Frobenius branch)**.** *For any branch and any $m \in S(p,5)$ with $m \bmod 5 \in \langle p \bmod 5\rangle$: $m \in \langle p\rangle \bmod M$. In particular, in the quartic branch ($p \equiv \pm 2 \bmod 5$), $S(p,5) = \langle p\rangle \bmod M$ unconditionally.*

Sketch. Multiply by a power of p to reduce to $m' \equiv 1 \pmod 5$; then the $T5$ rows give $(m'-1)e_a \equiv 0 \pmod N$ for *all* a, so $T_a \mid m'-1$ for the row orders T_a, hence $\mathrm{lcm}_a T_a = T \mid m'-1$, and $M = \mathrm{lcm}(T,5) \mid m'-1$. (Care is needed: distinct Frobenius orbits can have distinct row orders; the lcm over all rows repairs this.) □

The order-row interface needed below is kernel-checked independently of the full set equality in Theorem 6.2. The declaration `localS5_modEq_frobenius_power` compares a literal local row with the matching Frobenius-power row and proves congruence modulo $\mathrm{ord}(\zeta_5 - 1)$. Combining it with the already determined residue modulo 5, `quarticOrderRow_of_local` obtains the congruence modulo $M = \mathrm{lcm}(\mathrm{ord}(\zeta_5 - 1), 5)$, and `quarticOrderRows_of_skeleton` supplies it simultaneously at every prime factor. Finally, `inertQuotient_residue_determined` proves which quotient residue occurs, and a companion kernel theorem assembles this fact across all factors. Together they give the precise labelling $j_i = 0$ for $p_i \equiv \nu \pmod 5$ and $j_i = 2$ otherwise. These declarations prove necessity of the rows used in Theorem 6.3. The corresponding `ResultantTrap` declarations continue either literal final row to q^4-divisibility of its resultant, prove that the two fixed resultants are nonzero, and prove the closed estimate $q^4 \le |\operatorname{Res}| \le 16\cdot 5^A$. `ExplicitFiber` assembles the correct terminal branch at every skeleton factor. These declarations do not prove compatible-prefix CRT sufficiency or emptiness of any terminal fiber; the exact declaration names are listed in Section 10.

Theorem 6.3 (explicit $k = 3$ system)**.** *Under H4, a three-factor counterexample $n = p_1p_2p_3$ (all $p_i \equiv \pm 2 \bmod 5$) satisfies exactly*

$$n/p_i \equiv p_i^{j_i} \pmod{M_i}, \qquad j_i = \begin{cases} 0 & p_i \equiv \nu, \\ 2 & p_i \equiv -\nu, \end{cases} \qquad M_i = \operatorname{lcm}(T_i, 5),$$

with $T_i = \operatorname{ord}(\zeta_5 - 1)$ in $\mathbb{F}_{p_i^4}^\times$. Moreover $\operatorname{ord}_{p_i}(5) \mid n/p_i - 1$ for every i (base-5 Korselt condition shifted to n/p), and $M_i \le \max(n/p_i - 1, |n/p_i - p_i^2|)$.

The displayed branch-sensitive bound is now kernel-checked; its companion kernel theorem gives the universal exclusion when $n/p_i < p_i^2 \le M_i$. The declaration names are recorded in the public claim-status table. These are universal consequences of the literal row, not claims that the residual finite cases are empty.

The determination of the exponents uses Theorem 6.1(a) and the four-element lemma $p^4 \equiv 1 \pmod M$; necessity uses Theorem 6.2. The base-5 condition follows from the norm lemma below. A CRT-combinable Korselt system for this congruence, in the ring $\mathbb{Z}_n[X]/(X^5 - 1)$, was first set up by Váňa [11]; the present version localizes it to the field component, proves its necessity on the minimal skeleton, and determines the exponents.

7. The exact order decomposition

Let $p \equiv \pm 2 \pmod 5$ (quartic branch), $\zeta = \zeta_5 \in \mathbb{F}_{p^4}$, $T_p = \operatorname{ord}(\zeta - 1)$.

Lemma 7.1 (norm lemma)**.** $(\zeta - 1)^{1+p+p^2+p^3} = 5$ *in $\mathbb{F}_p^\times$; hence $\operatorname{ord}_p(5) \mid T_p$.*

Theorem 7.2 (exact decomposition)**.** *For odd p: $u_1^{p^2-1} = -\zeta^{-1}$ has order 10, whence $T_p \mid 10(p^2 - 1)$, $v_2(T_p) = 1 + v_2(p^2 - 1)$, $v_5(T_p) = 1$, and*

$$T_p = 2^{\,b_p+1} \cdot 5 \cdot d_{p,+} \cdot d_{p,5}, \qquad b_p = v_2(p^2 - 1),$$

with coprime odd parts $d_{p,+} \mid (p + 1)_{\mathrm{odd}}$ and $d_{p,5} \mid \operatorname{ord}_p(5)_{\mathrm{odd}}$.

Both the conjugation identity $u_1^{p^2-1} = -\zeta^{-1}$ and the resulting bound $T_p \mid 10(p^2 - 1)$ already appear in the Lenstra–Pomerance notes [10], and a ring version ($\rho \mid 10(\lambda^2-1)$ modulo $X^5 - 1$) in [11], Lemma 3.4: we claim no novelty for them. The exact 2-adic law is kernel-checked by `localCyclotomicUnit_order_factorization_two` and `quarticOrderModulus_factorization_two`. The coprime odd-tail decomposition and the Kummer interpretation below remain paper-level. None of these refinements was found in the targeted search; their priority remains provisional. The binary consequence for the Korselt rows is a single bit:

Corollary 7.3 (binary rows)**.** *Row i of the system implies $n/p_i \equiv 1+\eta_i 2^{b_i} \pmod{2^{b_i+1}}$, with $\eta_i \in \{0, 1\}$ according to the pure/twisted case. On census data this bit accounts for $\sim$90% of all row failures. However, the associated "at most one bit can hold" candidate theorem is false:*

an exhaustive enumeration modulo 2^7 *over the skeleton finds* $96+18{,}720$ *configurations (out of* 2^{21}*) where two, respectively three, bits hold simultaneously. Any universal incompatibility must therefore involve the odd tails. The later exact-ray corollary makes this binary compatibility completely explicit in the kernel; it is a classification of the binary projection, not an additional sieve after the full CRT triangle.*

8. Kummer identification of the tail

Theorem 8.1 (tail identification)**.** *Let* ℓ *be an odd prime with* $\ell \mid p+1$*, and let* $e_\ell = \max\{0 \leq e \leq v_\ell(p+1) : \zeta - 1 \in (\mathbb{F}_{p^4}^\times)^{\ell^e}\}$ *(Kummer depths are always truncated at the* ℓ*-adic valuation of the ambient group order: without the bound the maximum need not exist, e.g. every element of order coprime to* ℓ *is an* ℓ^e*-th power for all* e*). Then*

$$v_\ell(d_{p,+}) = v_\ell(p+1) - e_\ell, \qquad \textit{and consequently} \qquad q_p := \frac{(p+1)_{\mathrm{odd}}}{d_{p,+}} = \prod_{\ell \mid p+1,\ \ell \text{ odd}} \ell^{e_\ell}.$$

The tail quotient is exactly the Kummer defect of $\zeta - 1$ *in* $\mathbb{F}_{p^4}$*.*

Thus all local degeneracies classified in this paper are power-residue phenomena of canonical cyclotomic elements: at $k = 1$, the elements 5 and ε as 5^s-th powers (golden tower); at the $k = 3$ tail, $\zeta - 1$ as an ℓ^e-th power. One mechanism, three incarnations. (Whether the still-open order-4 branch obeys the same mechanism is precisely Hypothesis H4 territory, and we do not claim it.) On census data, $q_p = 1$ for $\sim$90% of the relevant rows (so the local condition is generically the full odd (+1)-side condition), and $q_p > 1$ is governed by small ℓ with the expected frequency $\sim \sum_\ell 1/(\ell(\ell-1))$.

9. The derived Lenstra–Pomerance mechanism

Since $d_{i,+} \mid p_i + 1$, we have $p_i^2 \equiv 1 \pmod{d_{i,+}}$: the pure and twisted congruences coincide on the tail, and the three-factor system collapses, after the binary filter and absorption of the factor 5, to the cyclic tail system

$$d_{1,+} \mid p_2p_3 - 1, \qquad d_{2,+} \mid p_1p_3 - 1, \qquad d_{3,+} \mid p_1p_2 - 1.$$

These are precisely the conditions $p_i + 1 \mid n + 1$ of the Lenstra–Pomerance construction [10], weakened to the structural divisor $d_{i,+}$ of $(p_i + 1)_{\mathrm{odd}}$, and *derived* as necessary conditions on the minimal skeleton rather than imposed as sufficient ones. Theorem 8.1 states exactly when the weakening is strict: when $\zeta - 1$ has a nontrivial Kummer defect. This gives a first-principles explanation, at the level of necessity, for why the (+1)-side of the Lenstra–Pomerance mechanism is the operative wall of Agrawal's congruence in the minimal composite case. We have not found this derivation in the targeted literature search, but make no absolute priority claim.

In fact the decomposition of Theorem 7.2 organizes the complete local row into *four checks with distinct roles*: the 2-adic bit; the factor 5 (absorbed by the skeleton); the (+1)-*jaw* $d_{p,+} \mid (p+1)_{\mathrm{odd}}$; and the (-1)-*jaw* $d_{p,5} \mid \mathrm{ord}_p(5)_{\mathrm{odd}}$, which lives on the $p - 1$ side since $\mathrm{ord}_p(5) \mid p - 1$. The local structure of Agrawal's congruence thus *internally generates both sides* $p \mp 1$ of the Lenstra–Pomerance mechanism, as the two coprime odd components of the order of $\zeta - 1$. Mirroring the tail defect, define the (-1)-side defect $q_{p,-} = \mathrm{ord}_p(5)_{\mathrm{odd}}/d_{p,5}$. The question closes in the strongest possible form: the (-1)-jaw carries no defect of its own; the odd $(p-1)$-component of T_p descends losslessly through the cyclotomic norm.

Theorem 9.1 (norm transfer for the odd $(p-1)$-component)**.** *Let $p \equiv \pm 2 \pmod 5$ be an odd prime and $\ell \neq 5$ an odd prime with $\ell \mid p-1$. Then*

$$v_\ell(\mathrm{ord}_p(5)) = v_\ell(T_p) = v_\ell(p-1) - f_\ell,$$

where f_ℓ is the Kummer depth of 5 in $\mathbb{F}_p^\times$ at ℓ (the largest $f \leq v_\ell(p-1)$ with $5 \in (\mathbb{F}_p^\times)^{\ell^f}$; truncation is essential: e.g. for $p = 13$, $\ell = 3$, the element 5 is a 3^e-th power for every e, and the intended value is $f_3 = 1$); moreover the Kummer depth of $\zeta - 1$ in $\mathbb{F}_{p^4}^\times$ at ℓ equals f_ℓ. Consequently $d_{p,5} = \mathrm{ord}_p(5)_{\mathrm{odd}}$ exactly, i.e. $q_{p,-} = 1$ identically.

Proof. By the norm lemma (Lemma 7.1), $5 = u^S$ with $u = \zeta - 1$ and $S = 1 + p + p^2 + p^3 = (p^4-1)/(p-1)$, so in the cyclic group $\mathbb{F}_{p^4}^\times$ we have the two exact identities

$$\mathrm{ord}_p(5) = \frac{T_p}{\gcd(T_p, S)}, \qquad T_p = \mathrm{ord}_p(5)\gcd(T_p, S).$$

For $\ell \mid p-1$: $S \equiv 4 \pmod \ell$, and ℓ is odd, so $\ell \nmid S$ and $v_\ell(\mathrm{ord}_p 5) = v_\ell(T_p)$. Since $\ell \nmid p+1$, $v_\ell(d_{p,+}) = 0$, so $v_\ell(d_{p,5}) = v_\ell(T_p) = v_\ell(\mathrm{ord}_p 5)$; at odd primes dividing $p+1$ both $d_{p,5}$ and $\mathrm{ord}_p(5)$ have trivial part, and $T_p \mid 10(p^2-1)$ excludes all remaining primes: hence $d_{p,5} = \mathrm{ord}_p(5)_{\mathrm{odd}}$. The depth formula in $\mathbb{F}_p^\times$ gives $v_\ell(\mathrm{ord}_p 5) = v_\ell(p-1) - f_\ell$; the depth formula in $\mathbb{F}_{p^4}^\times$ gives $v_\ell(T_p) = v_\ell(p^4-1) - e_\ell = v_\ell(p-1) - e_\ell$, using $v_\ell(p^4-1) = v_\ell(p-1)$ for odd $\ell \mid p-1$; comparing, $e_\ell = f_\ell$. (One inequality is conceptual: the norm maps ℓ-power classes to ℓ-power classes, so $f_\ell \geq e_\ell$; the order computation forces equality.) ☐

Verified with zero violations on all 1,138 primes $p \equiv \pm 2 \pmod 5$, $p < 20{,}000$ (depth identity $e_\ell = f_\ell$ checked directly for $p < 3{,}000$).

Remark 9.2 (domain of the norm-transfer argument)*.* The argument is specific to odd $\ell \mid p-1$ in the quartic branch: at $\ell = 2$ the exponent S is even and the 2-adic component obeys the separate exact law $v_2(T_p) = 1 + v_2(p^2-1)$; $\ell = 5$ never divides $p-1$ here; and outside the branch $p \equiv \pm 2 \pmod 5$ the Frobenius orbit, hence the norm identity itself, changes.

Corollary 9.3 (complete determination of T_p)**.**

$$T_p = 2^{\,1+v_2(p^2-1)} \cdot 5 \cdot d_{p,+} \cdot \mathrm{ord}_p(5)_{\mathrm{odd}},$$

and, expanding both odd components through their Kummer defects,

$$T_p = 2^{\,1+v_2(p^2-1)} \cdot 5 \cdot \frac{(p+1)_{\mathrm{odd}}}{q_{p,+}} \cdot \prod_{\ell \mid p-1,\ \ell \text{ odd}} \ell^{\,v_\ell(p-1) - f_\ell},$$

with $q_{p,+} = \prod \ell^{e_\ell}$ the Kummer defect of Theorem 8.1. The order of $\zeta - 1$ is completely determined by the Kummer defect vectors of the canonical pair: 5 *downstairs on the* (-1)*-side,* $\zeta - 1$ *upstairs on the* $(+1)$*-side.*

The picture is thus not symmetric but *completely asymmetric*, with the norm behaving in opposite ways on the two sides. Factor the norm exponent:

$$S = 1 + p + p^2 + p^3 = (p+1)(p^2+1).$$

For odd $\ell \mid p-1$ we have $\ell \nmid S$: the norm is invertible on the ℓ-Sylow and all information descends losslessly to $5 = \mathrm{N}(\zeta - 1) \in \mathbb{F}_p^\times$; the side is *norm-visible*. For odd $\ell \mid p+1$ we have $v_\ell(S) = v_\ell(p+1)$: the norm annihilates precisely the information living on that side; the side is *norm-invisible* and must be read directly on the residuosity of $\zeta - 1$ upstairs; accordingly its ℓ-Sylow lies in $\mathbb{F}_{p^2}^\times$ and the defect $q_{p,+}$ is genuinely quadratic. In summary: *the exact order* *decomposition internally derives the two Lenstra–Pomerance sides; the $p-1$ side transferred losslessly by the norm 5, the $p+1$ side governed by a genuine Kummer defect of the quartic cyclotomic element.*

In particular, the odd content of the (-1)-jaw condition in the three-factor system is exactly the odd part of the base-5 Korselt condition $\mathrm{ord}_{p_i}(5) \mid n/p_i - 1$ of Theorem 6.3. This proves the *structural complementarity* of the two experimental populations of Section 11: a candidate built on the base-5 condition has the (-1)-jaw closed by construction, so only the $(+1)$-jaw and the bit can eliminate it; dually, a seed built on the tails can only be eliminated by the (-1)-jaw and the bit. The observed mortality frequencies are computational evidence whose meaning the theorem explains; they are not themselves universal statements.

10. Finite fibers for the third prime

The two-row analysis of Section 9 leaves a single question per compatible pair: which third primes q satisfy their own complete row. This fiber is finite, effectively bounded, and enumerable.

Theorem 10.1 (finite fibers: Frobenius elimination and resultant trapping)**.** *Let (p_i, p_j) be a compatible pair, $A = p_i p_j - 1$, and let $q \equiv \pm 2 \pmod 5$ be a prime, inert in $K = \mathbb{Q}(\zeta_5)$, whose complete row holds in the triple (p_i, p_j, q). Write $u = \zeta_5 - 1$. Pure case ($j_q = 0$; the skeleton forces $10 \mid A$): then $u^A \equiv 1 \pmod q$, so q divides $\alpha_A^- = u^A - 1 \neq 0$ and*

$$q^4 \mid N_{K/\mathbb{Q}}(\alpha_A^-) = \mathrm{Res}\big(\Phi_5(X),\, (X-1)^A - 1\big).$$

Twisted case ($j_q = 2$; the skeleton forces $A \equiv 8 \bmod 10$): since $q^2 \equiv -1 \pmod 5$, Frobenius gives $u^{q^2} = \zeta_5^{-1} - 1 = -\zeta_5^{-1} u$, so the row $u^{A+1} = u^{q^2}$ is equivalent to $\zeta_5 u^A + 1 \equiv 0 \pmod q$, and

$$q^4 \mid N_{K/\mathbb{Q}}(\zeta_5 u^A + 1) = \mathrm{Res}\big(\Phi_5(X),\, X(X-1)^A + 1\big).$$

Both algebraic integers are nonzero; their vanishing would force $5^A = N(u)^A = 1$; and $|N| \le 16 \cdot 5^A$; since q is inert, (q) is a prime ideal of norm q^4, whence the q^4-divisibility and the effective bound

$$q \le 2 \cdot 5^{A/4}.$$

The fiber of admissible third primes over a compatible pair is therefore finite and enumerable among the inert prime divisors of two explicit integers.

The row-to-fixed-nonzero-resultant part of this theorem is kernel-checked by

`phi5_irreducible_of_inert`, `prime_pow_natDegree_dvd_resultant_of_common_root`,
`pure_row_pow_four_dvd_resultant`, `twisted_row_pow_four_dvd_resultant`,
`pure_resultant_mod_five_eq_one`, `twisted_resultant_mod_five_eq_one`,
`pure_row_pow_four_le_resultant_natAbs`, `twisted_row_pow_four_le_resultant_natAbs`,
`pure_resultant_natAbs_le`, `twisted_resultant_natAbs_le`,
`pure_row_pow_four_le_sixteen_mul_five_pow`,
`twisted_row_pow_four_le_sixteen_mul_five_pow`,
`explicitResultantRows_of_skeleton`,
`squarefree_counterexample_explicit_resultant_dichotomy`,
`no_squarefree_counterexample_of_no_split_and_no_explicit_rows`.

The first declaration proves irreducibility in the inert branch. The generic theorem identifies the reduction of Φ_5 with the minimal polynomial of the common root, proves that the integral division remainder is coefficientwise divisible by q, and uses resultant homogeneity to extract $q^{\deg \Phi_5} = q^4$. The two `pow_four_dvd_resultant` declarations supply the displayed pure and twisted vanishing relations from a literal `LocalS5` row. The mod-5 declarations then use $\Phi_5 = (X-1)^4$ and evaluation at $X = 1$ to prove $\mathrm{Res} \equiv 1 \pmod 5$, while the final pair combines nonvanishing with divisibility to obtain $q^4 \le |\,\mathrm{Res}\,|$. The next four declarations evaluate the resultant over the four complex roots of Φ_5, prove that the product of the four distances $|\zeta - 1|$ is 5, and obtain the displayed closed bound $q^4 \le 16 \cdot 5^A$. The final two declarations show that every factor of the concrete quartic skeleton lands in the correct terminal resultant branch. Compatible-prefix CRT sufficiency and universal emptiness of the resulting fibers remain paper-level/open.

Lemma 10.2 (quadratic descent)**.** *Both norms are squares. With $t = \zeta_5 + \zeta_5^{-1} = (\sqrt{5}-1)/2$:*

$$U := u^{10} = -175 + 275\,t, \qquad V := \zeta_5 u^8 = 50 - 75\,t$$

lie in $\mathbb{Q}(\sqrt{5})$, with $\mathrm{Tr}(U) = -625$, $N(U) = 3125$, $\mathrm{Tr}(V) = 175$, $N(V) = 625$. Writing $A = 10h$ (pure) or $A = 10h + 8$ (twisted), we get $\alpha_A^- = U^h - 1$, $\alpha_A^{tw} = VU^h + 1$, so $N_{K/\mathbb{Q}}(\alpha) = N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\alpha)^2$ with

$$D_h = 3125^h - S_h + 1, \qquad E_h = 625 \cdot 3125^h + R_h + 1,$$

where $S_h = \mathrm{Tr}(U^h)$, $R_h = \mathrm{Tr}(VU^h)$ satisfy the order-two recurrence $x_{h+2} = -625\,x_{h+1} - 3125\,x_h$ with $S_0 = 2$, $S_1 = -625$, $R_0 = 175$, $R_1 = -106250$. For an inert prime q, the complete third row (binary bit included) is equivalent to $q^2 \mid D_h$ (pure), respectively $q^2 \mid E_h$ (twisted): indeed q is inert in $\mathbb{Q}(\sqrt{5})$ as well, so there is a unique prime above q, of norm q^2, and $q^2 \mid N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\alpha) \iff \alpha \equiv 0 \pmod q$. For $A = 480$ the integer to factor has 168 digits instead of 336.

All identities, recurrences and both equivalences were verified exactly (symbolic computation in $\mathbb{Z}[\zeta_5]$; equivalence against the field row on $120 + 120$ random cases; the twisted witness of Section 11 as regression test). Cyclic-resultant recurrences of this kind are classical objects [26]; the reduction from Agrawal's row to this specific resultant trap, including the twisted branch after Frobenius elimination, was not found in the targeted search. We make no absolute priority claim.

Two further exact statements make the fibers effectively computable.

Lemma 10.3 (even valuation at inert primes)**.** *Let $\beta \in \mathcal{O}_{\mathbb{Q}(\sqrt{5})} \setminus \{0\}$ and $q \neq 5$ a prime inert in $\mathbb{Q}(\sqrt{5})$. Then $v_q(N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\beta)) = 2\,v_{(q)}(\beta)$: an inert prime enters the norm with even exponent (cost at least q^2), while a split prime can enter with exponent one (cost q). This is the deterministic seed of the observed factor pattern: the observed factorizations are split-dominated, consistently with the double valuation cost. (The lemma proves the q^2 cost, not yet a universal distribution statement.)*

Lemma 10.4 (unified cyclotomic engine)**.** *$V^5 = (\zeta_5 u^8)^5 = u^{40} = U^4$; since $5 \nmid q^2 - 1$ for q inert, fifth power is an automorphism of $\mathbb{F}_{q^2}^\times$, so in the twisted branch (with $k = 5h+4 = A/2$)*

$$VU^h = -1 \iff U^k = -1,$$

and the twisted fiber integer also splits cyclotomically: $N(U^k + 1) = \prod_{d \mid 2k,\ d \nmid k} N(\Phi_d(U))$. Both branches are thus driven by one engine, $U^h = 1$ (pieces $d \mid h$) and $U^k = -1$ (pieces $d \mid 2k$, $d \nmid k$), whose pieces are norms of cyclotomic values, small enough to factor.

Theorem 10.5 (two-row transport)**.** *Let $P = pr$ and suppose the final row and the row at p, after multiplication by p, have the forms*

$$P \equiv c \pmod{T_q}, \qquad Pq \equiv p^{j_p+1} \pmod{T_p}.$$

Then

$$\boxed{cq \equiv p^{j_p+1} \pmod{\gcd(T_p, T_q)}}.$$

Thus the pure and twisted branches force, respectively,

$$q \equiv p^{j_p+1}, \qquad q^3 \equiv p^{j_p+1} \pmod{\gcd(T_p, T_q)}.$$

Moreover, writing the final row as $P = 1 + hT_q$ in the pure branch or $P = q^2 - hT_q$ in the twisted branch, the complete lift at p is the linear congruence

$$(qT_q)h \equiv p^{j_p+1} - q \pmod{T_p} \quad \textit{(pure)},$$
$$(qT_q)h \equiv q^3 - p^{j_p+1} \pmod{T_p} \quad \textit{(twisted)}.$$

Proof. Reduce both input rows modulo $\gcd(T_p, T_q)$ and eliminate Pq. The two linear forms follow by substituting the displayed parameterizations of P. These are exact equivalences, not probabilistic sieves. □

Corollary 10.6 (three-row CRT triangle)**.** *If the second small row is*

$$Pq \equiv r^{j_r+1} \pmod{T_r},$$

then necessarily

$$\boxed{p^{j_p+1} \equiv r^{j_r+1} \pmod{\gcd(T_p, T_r)}.}$$

Proof. Reduce the two congruences for the same integer Pq modulo $\gcd(T_p, T_r)$. Together with the two instances of Theorem 10.5, this is the complete triangle of pairwise CRT compatibility. □

Corollary 10.7 (exact labelled dyadic rays)**.** *For each $x \in \{p, q, r\}$, put*

$$b_x = v_2(x^2 - 1), \qquad e_x = j_x + 1 \in \{1, 3\},$$

and let $\delta(1) = 0,\ \delta(3) = 1$. Every complete three-row triangle satisfies

$$b_p = b_q = b_r =: b$$

and

$$p + \delta(e_p)2^b \equiv q + \delta(e_q)2^b \equiv r + \delta(e_r)2^b \pmod{2^{b+1}}.$$

In particular $p \equiv q \equiv r \pmod{2^b}$.

Proof. Theorem 7.2 gives $2^{b_x+1} \mid T_x$. If two row targets are congruent modulo $\gcd(T_x, T_y)$, comparison one level above the smaller exact depth forces $b_x = b_y$. At the common depth,

$$x^1 \equiv x, \qquad x^3 \equiv x + 2^b \pmod{2^{b+1}},$$

because $x^2 \equiv 1 + 2^b \pmod{2^{b+1}}$. Applying this to the two transported sides of the triangle proves the displayed chain. The complete literal-row statement is kernel-checked as `quarticThreeRows_exact_dyadicRays`; the intermediate declarations include `square_modEq_one_add_exactTwoPower` and `quarticTriangle_exact_dyadicRays`. □

Corollary 10.8 (middle-prime defect cascade)**.** *Let $p < r < q$, and write*

$$T_r I_r = 10(r^2 - 1)$$

for the exact defect identity at r. Every three-factor candidate satisfies

$$\boxed{I_r \geq 11 \quad \textit{or} \quad r^2 < pq.}$$

Proof. If $I_r \le 9$, then $T_r > r^2$. The universal local-row size bound $T_r \le \max(r^2, pq)$ therefore forces $r^2 < pq$. Thus the middle prime must itself be in the exceptional Kummer range, or the final prime must realize the large gap $q > r^2/p$. □

Corollary 10.9 (quantized local gap)**.** *Let $P_x = n/x$. A local row at x forces*

$$j_x = 0: \quad T_x \le P_x - 1, \qquad j_x = 2: \quad T_x \le |P_x - x^2|$$

whenever $P_x \ne x^2$. In particular, in the twisted branch no candidate lies strictly inside the open interval of radius T_x around x^2.

Proof. The two differences are nonzero multiples of T_x, hence have size at least T_x. □

Corollary 10.10 (branch-independent odd shadow)**.** *Remove the 2- and 5-parts from the local order:*

$$D_x = \frac{T_x}{2^{v_2(T_x)}5^{v_5(T_x)}}.$$

In the inert branch, $D_x \mid x^2 - 1$, and both local residues collapse to

$$P_x \equiv 1 \pmod{D_x}.$$

Thus a three-factor candidate satisfies

$$\boxed{D_p \mid rq - 1, \qquad D_r \mid pq - 1, \qquad D_q \mid pr - 1.} \tag{7.1}$$

Moreover,

$$\gcd(D_p, D_r) \mid r - p$$

and cyclically for the other two pairs. For $p < r < q$, equivalently,

$$\boxed{\gcd(D_p, D_r) \le r - p, \quad \gcd(D_p, D_q) \le q - p, \quad \gcd(D_r, D_q) \le q - r.}$$

Thus any reversed strict inequality is an exact certificate excluding the three-row triangle.

Proof. The structural order bound gives $D_x \mid x^2 - 1$, so the residues 1 and x^2 agree modulo D_x. For the last assertion, reduce two relations in (7.1) modulo $\gcd(D_p, D_r)$. The common factor q is a unit because $rq \equiv 1$; cancellation gives $p \equiv r$. The concrete definition of D_x, the deduction $T_x \mid 10(x^2 - 1) \Rightarrow D_x \mid x^2 - 1$, the complete three-pair incidence matrix and the strict rejection criterion are kernel-checked as `quarticOddTail_dvd_sq_sub_one`, `quarticOddTail_incidenceTriangle`, `quarticOddTail_incidenceBounds`, and `quarticOddTail_oversize_exclusion`. □

Corollary 10.11 (exact norm-visible two-jaw decomposition)**.** *For a good inert prime x, put $u = \zeta_5 - 1$ and define*

$$D_{x,-} = \mathrm{ordCompl}_2(\mathrm{ord}_x(5)), \qquad D_{x,+} = \gcd(D_x, x + 1).$$

Then

$$\boxed{u^{\,1+x+x^2+x^3} = 5, \qquad D_{x,-} = \gcd(D_x, x - 1),}$$

and the complete odd tail factors as

$$\boxed{D_x = D_{x,-}D_{x,+}, \qquad D_{x,-} \mid x - 1, \qquad D_{x,+} \mid x + 1, \qquad \gcd(D_{x,-}, D_{x,+}) = 1.}$$

If two compatible rows are attached to $p \le r$, every common odd prime has a consistent sign:

$$\ell \mid D_{p,-}, D_{r,+} \Longrightarrow \ell \mid 2, \qquad \ell \mid D_{p,+}, D_{r,-} \Longrightarrow \ell \mid 2.$$

In fact the complete common tail has the exact signed factorization

$$\boxed{\gcd(D_p, D_r) = \gcd(D_{p,-}, D_{r,-})\ \gcd(D_{p,+}, D_{r,+}).}$$

Consequently both same-sign overlaps occur simultaneously in the gap:

$$\gcd(D_{p,-}, D_{r,-})\ \gcd(D_{p,+}, D_{r,+}) \mid r - p.$$

Proof. The norm exponent

$$1 + x + x^2 + x^3 = (x + 1)(x^2 + 1)$$

runs through the four Frobenius conjugates of u, whose product is $\Phi_5(1) = 5$. Every odd divisor of $x-1$ is coprime to this exponent, so the norm cannot erase any odd $x-1$-component of $\mathrm{ord}(u)$. This proves the formula for $D_{x,-}$. Since $D_x \mid (x-1)(x+1)$ and D_x is odd, Euclidean cancellation gives the complementary factor and coprimality.

For the cross-sign assertion, Corollary 10.10 gives $\gcd(D_p, D_r) \mid r - p$. A divisor of $D_{p,-}$ and $D_{r,+}$ therefore divides $p - 1$, $r + 1$, and $r - p$, hence divides 2; the reverse orientation is identical. The literal norm identity, exact order quotient and norm-kernel factorization are kernel-checked in `QuarticNormJaw.lean`. Cancelling the two cross gcds gives the exact same-sign factorization and the signed incidence triangle, kernel-checked in `QuarticSignedIncidence.lean`. □

Remark 10.12 (the exact remaining odd wall). Corollaries 10.10 and 10.11 are deterministic sieves, but they are not a universal contradiction. The exact factorization forces a consistent sign on every shared odd prime, but it does not force two rows to share an odd prime or force a sign change. Nor does the current theory force any one of the three shared gcds to exceed its corresponding prime gap. A universal closure therefore still requires either a class-specific lower bound or mixing theorem for the joint jaws, or a theorem emptying every explicit terminal resultant fiber.

The chain of reductions supports a correctness theorem for machine certificates, which a referee can audit once instead of trusting each output separately.

Theorem 10.13 (certificate correctness)**.** *Suppose an implementation, for a compatible pair in a fixed branch, (i) computes the cyclotomic pieces $N(\Phi_d(U))$ of the fiber integer and verifies that their product reconstructs it exactly (sign and multiplicities included); (ii) factors every piece completely; (iii) certifies every factor prime by a proven primality test; (iv) checks even valuation at every inert factor; and (v) applies the inertia and CRT-class filters to the complete factor list, finding no survivor. Then the pair admits no third prime in that branch: no counterexample p_ip_jq exists for any prime q.*

Proof sketch. Necessity of the third row (Theorems 6.2 and 6.3) plus the resultant trap (Theorem 10.1) place every candidate q among the inert prime divisors of the fiber integer; the quadratic-descent equivalence (Lemma 10.2) makes membership exact; steps (i)–(iii) guarantee the divisor list is complete and correct, and step (v) is exhaustive over it. (Proven primality matters: composite-factoring routines may return strong pseudoprimes unless primality is certified separately.) □

As a first application we record the first *complete empty-fiber certificate*: for the pair $(13, 37)$ ($A = 480$, both residue classes pure), the cyclotomic splitting of D_{48} into ten pieces (largest: 56 digits) was fully factored, all 32 prime factors certified prime, the product reconstructed exactly, and *no* inert prime beyond $\{2, 3, 7\}$ divides D_{48} at all; in particular none lies in the two admissible CRT classes. Hence *no prime q whatsoever makes $13 \cdot 37 \cdot q$ a counterexample at $r = 5$*: the earlier "zero survivors up to 10^9" becomes an absolute statement for this pair.

Two further reductions sharpen the fiber analysis. First, the *level theorem with CRT sieve*: if q is inert, $q \nmid d$ and $q \mid N(\Phi_d(U))$, then $\mathrm{ord}_{\mathbb{F}_{q^2}^\times}(U) = d$, hence $d \mid q^2 - 1$; if moreover $q \equiv r \pmod{M}$ (the two-row class), then necessarily $r^2 \equiv 1 \pmod{\gcd(d, M)}$; a purely arithmetic condition that can eliminate entire cyclotomic pieces before any factoring (for the pair $(13, 37)$ it eliminates exactly the levels 16 and 48, i.e. the two largest pieces). In the non-primitive case $q \mid d$ the level drops to $d/q^{v_q(d)}$ and the cost in the norm is *exactly* q^2 (local valuation one, by the lifting-the-exponent lemma); at the primitive level the q^2 cost is a minimum (Wieferich-type excess is not excluded). Second, on the counting side, $B(A) \le 1$: the compatible pair is determined by A, since $A + 1 = p_i p_j$ factors uniquely; so the sum over pairs is a sum over semiprime values of $A+1$, with an unconditional aggregate bound $O(X \log\log X/\log X)$. The equidistribution hypothesis needed for global finiteness can then be stated quantitatively (UED): over dyadic blocks, $N_j \le C\, M_j + E_j$ with $\sum_j M_j$, $\sum_j E_j < \infty$, uniformly in the division fields and CRT moduli involved; which deterministically implies that the number of three-factor counterexamples is finite. UED with summable errors is precisely what remains on this route.

We note finally that $D_h = N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(U^h - 1)$ is a Lehmer–Pierce sequence for the quadratic integer U; the literature on primitive divisors of Lucas, Lehmer and Lehmer–Pierce sequences (see e.g. [23, 24, 27]) may provide structural information on the divisors that our inertia filter then screens, though it does not replace the filter itself.

A fundamental caution: finiteness of each fiber does *not* bound the total number of three-factor counterexamples, since the compatible pairs may be infinite in number. The global $k = 3$ problem thus decomposes as

$$\textit{uniform control of compatible pairs} \; + \; \textit{factorization of the finite fibers } D_h, E_h,$$

so the pointwise algebraic wall is removed while the global union remains open.

The machine extends verbatim to every odd number of factors.

Theorem 10.14 (general-k reduction)**.** *Let $n = p_1 \cdots p_{k-1} q$ be squarefree in the quartic skeleton ($k \ge 3$ odd, all factors $\equiv \pm 2 \bmod 5$, $\nu = n \bmod 5 \in \{2, 3\}$), and put $P = p_1 \cdots p_{k-1}$, $A = P - 1$. (i) The first $k - 1$ rows are simultaneously solvable in q iff $\gcd(P/p_i, T_i) = 1$ for all i and the residues $a_i \equiv p_i^{j_i}(P/p_i)^{-1} \pmod{T_i}$ are pairwise compatible (together with the mod-5 class); the solutions then form a single reduced class $q \equiv r \pmod{M}$, $M = \mathrm{lcm}(5, T_1, \ldots, T_{k-1})$. For $k = 3$ these conditions are exactly the pair-compatibility conditions above; for $k > 3$ they are genuinely $(k-1)$-ary. (ii) For every compatible $(k-1)$-tuple, the admissible last primes q lie in the same two resultant traps, with $A = P - 1$: q^4 divides $\mathrm{Res}(\Phi_5, (X-1)^A - 1)$ (pure) or $\mathrm{Res}(\Phi_5, X(X-1)^A + 1)$ (twisted), and $q \le 2 \cdot 5^{A/4}$: each fiber is finite and enumerable. (iii) Under H4, Agrawal's conjecture for $r = 5$ restricted to squarefree integers is equivalent to: no compatible $(k-1)$-tuple, for any odd $k \ge 3$, has a fiber prime passing its own complete row.*

Theorem 10.15 (compatible prefixes at arbitrary length)**.** *For every length r, residue $\nu \in \{2, 3\}$ and bound B, there are distinct primes $p_1, \ldots, p_r > B$, all congruent to $\nu \pmod{5}$, whose first r rows form a compatible CRT prefix.*

Proof idea. If $T_i = \mathrm{ord}_{\mathbb{F}_{p_i^4}^\times}(\zeta_5 - 1)$, compatibility is equivalent to $p_j \nmid T_i$ for $i \ne j$ and to the consistency of the single system $p_i^{j_i+1} \pmod{T_i}$. Given a compatible tuple, choose a new prime in the common reduced class modulo $\mathrm{lcm}(T_i)$, while avoiding the at most four fourth roots of unity modulo each old prime. The ordinary CRT and Dirichlet's theorem provide arbitrarily large such primes. Since $T_{\text{new}} \mid p_{\text{new}}^4 - 1$, the avoided classes give the reverse coprimalities as well. □

Thus there is no collapse obtained merely by comparing the number of pairwise congruences with the number of variables. Any global exclusion must use the final fiber row. The exact reduction is

$$\text{tuple} \longrightarrow \text{CRT} \longrightarrow \text{resultant} \longrightarrow \text{exact filter}.$$

Global case	Available result	Status
$s = 2$	excluded by the structure theorem	conditional on H4
even $s \geq 4$	excluded by the parity/order-4 structure	conditional on H4
$s = 3$, stated box	no survivor in the exact finite searches	certified computation under H4
$s = 3$, seven pairs	fibers empty without a bound on q	exact certificates
$s = 3$, remaining pairs	exact CRT and finite terminal fiber	reduction theorem; open union
odd $s \geq 5$	compatible prefix plus finite terminal fiber	reduction theorem; open union

11. Certified computations

All computations are exact (finite-field arithmetic, no floating point) and scripted; every eliminated candidate carries an inspectable certificate (the failing row and the failing modulus). Sanity identities whose truth is a theorem (Frobenius row transport, solvability of $t = p^j$) are asserted in every run.

Computational verification 11.1 (certified finite region, under H4). Assume H4. There is no squarefree counterexample $n = p_1p_2p_3$ at $r = 5$ with $7 \leq p_1 < p_2 < 30{,}000$ and $p_3 < 10^9$; the boundary $p_1 \in \{2, 3\}$ is excluded for $p_3 < 2 \cdot 10^8$.

Computational verification 11.2 (largest-prime census, under H4). There is no squarefree three-factor counterexample at $r = 5$ whose largest prime factor is $q < 10^6$. The replay enumerates the entire all-inert branch: among 39,286 inert primes it finds 76,530 residual final-row products and 415 admissible semiprimes. The multiplier-free transport of Theorem 10.5 eliminates 391. Of the remaining 24, exactly 17 pass the complete CRT triangle of Corollary 10.6, while exactly one passes both local-row size bounds; these two sets are disjoint. Thus the box is already empty before solving the lift. As a redundant check, for every one of the 24 the canonical residue required by the first reduced linear lift lies strictly above the exact multiplier interval.

The H4 qualifier is essential: the replay exhausts the all-inert fiber branch, whereas a split local order-4 witness belongs to the other arm of the unconditional dichotomy.

The finite disjointness just used is not asserted universally. A targeted sharded falsifier extends the all-inert census to $q < 10^7$. It enumerates 332,441 inert final primes, 993,466 residual products, and 4,429 admissible semiprimes. Of these, 254 pass both transports, 194 pass the CRT triangle, 14 pass both local-row size bounds, and 10 pass both structural tests. All ten fail the exact small-prime rows. Thus there is no all-inert three-factor counterexample in this finite region (and, under H4, no three-factor counterexample there at all). A second replay applies Corollaries 10.9 and 10.10 to the ten frozen records: the quantized gap eliminates two, while the branch-independent odd shadows eliminate all ten. Thus this finite box does not require discrete-log reconstruction of the complete lifts. This adversarial extension isolates the remaining universal problem at the three simultaneous moving odd moduli in (7.1), together with the 2- and 5-adic locks, and prevents the initial finite pattern from being mistaken for a theorem.

Method: the base-5 sieve of Theorem 6.3 places p_3 in a single CRT progression for each admissible pair (p_1, p_2); 1,330,896 pairs produce 2,227 candidates (plus 1,053 and 369 in two earlier sweeps, and 2 on the boundary), each judged by the exact test $(\zeta - 1)^{n/p_i - p_i^{j_i}} = 1$ in $\mathbb{F}_{p_i^4}$; zero survive. Across the earlier base-5 population, the first row causes nearly every elimination. This is a distribution inside a finite census, not a universal obstruction. Indeed

$$(p_1, p_2, q) = (13, 37, 30517)$$

passes the complete rows at 13 and 37 and fails only the row at 30517; the residue of the failing row is $480 = p_1 p_2 - 1$. For $p_1 = 3$ one has $T_3 = 80$ and exactly $16 + 16$ admissible classes mod 80, so no "small-factor barrier" exists: the walls are the large factors' rows, as the census confirms.

Computational verification 11.3 (unbounded empty fibers). The exact level certificates prove that the fibers over

$$(13, 37),\ (17, 113),\ (13, 277),\ (23, 167),\ (7, 823),\ (83, 347),\ (463, 1327)$$

are empty for every possible last prime. The universal-level cache contains 76 completely reconstructed and prime-certified values $|N(\Phi_d(U))|$. On the frozen queue of 356 compatible pairs, 7 are certified empty, 349 remain open because at least one required level is unfactored, and no verified survivor occurs.

The two relevant manifests are

```
fibre_s42/MANIFEST.json
MANIFEST_RIPRODUCIBILITA_AGRAWAL.json.
```

They record the detector, required levels, primality proofs, exact reconstructions, original-field checks and SHA-256 hashes. The replay verifier recomputes all 76 level values and all 356 pair plans. Timeouts are open fibers, never negative results.

A complementary experiment conditions on the $(+1)$-side instead: a search for skeleton triples satisfying all three tail conditions $(p_i + 1)_{\mathrm{odd}} \mid p_j p_k - 1$ simultaneously ("seeds") finds them in abundance, so the universal incompatibility of the three tails is *false*. On a frozen corpus of 39 seeds (archived with the verdicts for exact reproducibility) the complete-row judgment gives: zero survivors; of the 116 failing rows, the binary bit blocks 102 and the (-1)-jaw d_5 blocks 111 (d_5 alone: 14; bit alone: 5; the tail and the factor 5: never, as forced by the seed construction and the skeleton). The two populations are thus eliminated by *complementary* jaws, exactly as the bilateral picture of Section 9 predicts. One seed, (7, 367, 28049407), passes the complete row at $p = 7$: the first witness of solvability of a complete row in the three-factor system (it shows the four checks of a row can be satisfied simultaneously; it does not suggest proximity to satisfying all three rows). A caveat on composition: the snapshot was taken while the seed enumeration was still at $p_1 = 7$, and Mersenne-type primes are privileged seed components since $(p + 1)_{\mathrm{odd}} = 1$ empties the tail; quantitative frequencies therefore require the completed census or stratified sampling.

12. Counting, and open problems

Let $M_s^{(4)}(X)$ count squarefree counterexamples in the quartic skeleton with exactly s prime factors. The terminal row gives a first quantitative theorem for the global union of fibers.

Theorem 12.1 (compression of the terminal-fiber wall)**.** *For every fixed $s \geq 3$ and $X \geq 256$,*

$$M_s^{(4)}(X) \leq C_s \frac{X}{\log X}\big(\log\log(2X)+2\big)^{s-2}, \tag{10.1}$$

where

$$C_s = (s-1)\left(\frac{\log 16}{2} + 2\log 5\right) + \frac{16s(s-1)}{(s-2)!\log 2}. \tag{10.2}$$

Under H4 the same bound holds for all squarefree counterexamples with s factors; for even s the count is zero.

Proof architecture. Write the largest prime as q, the product of the other $k = s-1$ primes as P, and split at $P = \sqrt{X}$. If $P \leq \sqrt{X}$, the two resultant traps satisfy

$$q^4 \mid \mathcal{R}_{P,\pm}, \qquad \log|\mathcal{R}_{P,\pm}| \leq \log 16 + (P-1)\log 5.$$

Since the largest prime of the prefix is at least $P^{1/k}$, each fiber has at most

$$\frac{k[\log 16 + (P-1)\log 5]}{2\log P}$$

primes across both branches. Summing over all integers $P \leq \sqrt{X}$ gives the first term in (10.2). If $P > \sqrt{X}$, ignore the final row and count k-prime prefixes in dyadic blocks. Choosing the largest prefix prime uniquely and using $\sum_{p\leq y} p^{-1} \leq \log\log y + 2$ gives

$$\#\{P \in [Z, 2Z)\} \leq \frac{4kZ}{(k-1)!\log Z}(\log\log(2Z)+2)^{k-1}.$$

The prime bound for $q \leq X/P$, the inequality $P < X^{k/(k+1)}$, and summation over the dyadic blocks yield the second term in (10.2). The proof is unconditional inside the quartic skeleton; only the final application to all branches uses H4. ☐

The unrestricted squarefree s-almost-prime count has one additional factor of $\log\log X$. Thus Theorem 12.1 gives a genuine compression, not vacuity. Heath-Brown's stronger bound for Carmichael numbers with three prime factors [4] uses the full Korselt divisibilities and does not transfer to the weaker order conditions here. Likewise the Alford–Granville–Pomerance construction [3] is an existence method, not an upper bound for this set. The explicit elementary estimates used above are in the range of Rosser–Schoenfeld [9].

We next state the Galois-theoretic density program for the tail anomalies.

Theorem 12.2 (Theorem A: density of tail anomalies at level ℓ)**.** *Let $\ell \neq 5$ be an odd prime, $K = \mathbb{Q}(\zeta_5)$, $K_0 = \mathbb{Q}(\zeta_5, \mu_\ell)$, and $\widetilde{K}_\ell = K_0\big((\zeta_5^a - 1)^{1/\ell} : a \in (\mathbb{Z}/5)^\times\big)$, which is Galois over $\mathbb{Q}$. (i) The classes of the four conjugates $u_a = \zeta_5^a - 1$ in $K^\times/(K^\times)^\ell$ span exactly the two-dimensional space $\langle [u_1], [\varepsilon]\rangle$: indeed $u_2 = (-\zeta_5^3\varepsilon)\, u_1$, $u_3 = (\zeta_5\varepsilon)\, u_1$, $u_4 = (-\zeta_5^4)\, u_1$, and roots of unity in μ_{10} are ℓ-th powers; independence follows from $v_\lambda(u_1) = 1$, $v_\lambda(\varepsilon) = 0$ at the unique prime λ above 5, together with $\mathcal{O}_K^\times = \mu_{10} \times \langle\varepsilon\rangle$. The rank persists over K_0 because $[K_0 : K] \mid \ell - 1$ is coprime to ℓ. Hence*

$$[\widetilde{K}_\ell : \mathbb{Q}] = 4(\ell-1)\,\ell^2.$$

(ii) On the Kummer space with basis $([u_1], [\varepsilon])$ the generator $\sigma : \zeta_5 \mapsto \zeta_5^2$ acts by $S = \left(\begin{smallmatrix} 1 & 0 \\ 1 & -1 \end{smallmatrix}\right)$, $S^2 = I$. For a lift $g = (w, h)$ with h of order 4 on K and inversion on μ_ℓ, the induced action on $W = \mathrm{Gal}(\widetilde{K}_\ell/K_0) \cong \mathbb{F}_\ell^2$ is $R = -S^T$, and

$$g^4 = \big(2(I + R)\, w,\ 1\big), \qquad 2(I + R)(a, b) = (-2b,\, 4b),$$

so g^4 fixes an ℓ-th root of u_1 iff $b = 0$ iff $g^4 = 1$. The class C_ℓ cut out by these conditions is stable under conjugation, with $|C_\ell| = 2\ell$. (iii) By Chebotarev, the density of primes $p \equiv \pm 2 \pmod 5$, $p \equiv -1 \pmod \ell$ such that $\zeta_5 - 1$ is an ℓ-th power in $\mathbb{F}_{p^4}$ equals

$$\delta_\ell = \frac{2\ell}{4(\ell-1)\ell^2} = \frac{1}{2\ell(\ell-1)}; \qquad \textit{conditioned on the two base congruences: exactly } \frac{1}{\ell}.$$

Numerical control: for $\ell = 3$, $1496/4495 = 0.3328$ against $1/3$ (and 0.083185 against $1/12$ unconditionally) over $p < 2 \cdot 10^5$; for $\ell = 7$, 0.1369 against $1/7$. The proof is the Galois–Chebotarev computation, not the finite frequencies.

The theorem refines to every depth, and the disjointness across primes ℓ is itself a theorem. At depth $e \geq 1$, with $\widetilde{K}_{\ell,e} = K(\mu_{\ell^e})\big(u_a^{1/\ell^e}\big)$, the Kummer rank remains 2 (same two directions; the ℓ-adic descent requires eliminating a root-of-unity factor, which is done explicitly), $[\widetilde{K}_{\ell,e} : \mathbb{Q}] = 4(\ell-1)\ell^{3e-1}$, and the density of depth-$\geq e$ anomalies is

$$\delta_{\ell,e} = \frac{1}{2(\ell-1)\,\ell^{2e-1}}, \qquad (\text{conditioned: } \ell^{-e});$$

the exact-depth law is $(\ell+1)/(2\ell^{2e+1})$.

Theorem 12.3 (fiber-product disjointness)**.** *Let $G_\ell = \mathrm{Gal}(\widetilde{K}_\ell/K)$. Then $[G_\ell, G_\ell] = W_\ell$: the abelianization is the cyclotomic part. Consequently, for distinct admissible primes, any common quotient of G_ℓ and G_m has trivial commutator (an ℓ-group and an m-group at once), hence is abelian, hence cyclotomic, and $K(\mu_\ell) \cap K(\mu_m) = K$ by conductors: so $\widetilde{K}_\ell \cap \widetilde{K}_m = K$, and by induction the compositum over any finite set $\mathcal{L}$ has Galois group $\prod_{\ell \in \mathcal{L}} G_\ell$ over K (the full fiber product over $\mathrm{Gal}(K/\mathbb{Q})$).*

Therefore, *unconditionally*, for every finite $\mathcal{L}$ the density of "anomalous for some $\ell \in \mathcal{L}$" in the quartic branch equals $1 - \prod_{\ell \in \mathcal{L}}(1 - \frac{1}{\ell(\ell-1)})$. The passage to the full Euler product requires a uniform tail bound (UKS, the uniform Kummer sieve: the upper density of $\bigcup_{\ell > Y} E_\ell$ tends to 0 with Y, preserving the Kummer factor $1/\ell$); an Artin–Hooley type problem; simple Brun–Titchmarsh on the progressions $p \equiv -1 \pmod \ell$ provably does not suffice on ranges with distinct exponents. This is where the minimal-field structure of the next theorem enters.

The uniform tail is now a theorem under a standard, explicitly named hypothesis. The key is a third Frobenius elimination, which identifies the tail events with order anomalies of the golden unit:

$$\begin{aligned} &(\zeta_5 - 1)^{(p-1)(p^2+1)} = \varepsilon^2 && \text{in } \mathbb{F}_{p^4}, \\ &p \in E_\ell \iff \varepsilon^{2(p+1)/\ell} = 1 \iff \varepsilon \in (\mathbb{F}_{p^2}^\times)^\ell && (\ell \mid p+1). \end{aligned}$$

Theorem 12.4 (tail density under GRH)**.** *Assume GRH for the Dedekind zeta functions of the explicit family $M_\ell = \mathbb{Q}(\sqrt{5}, \mu_\ell, \varepsilon^{1/\ell})$, $\ell \neq 2, 5$ prime. Then the set of primes in the quartic branch having at least one tail anomaly has relative natural density exactly*

$$\Delta = 1 - \prod_{\ell \neq 2,5} \Big(1 - \frac{1}{\ell(\ell-1)}\Big) = 1 - \frac{40\,\mathcal{A}}{19} = 0.212724602906942\ldots,$$

where $\mathcal{A}$ is Artin's constant; the absolute density among all primes is $\Delta/2$.

Proof architecture. *(i) Minimal field.* By the equivalence above, $p \in E_\ell \iff \mathrm{Frob}_p(M_\ell/\mathbb{Q}) = h$, where M_ℓ is the *minimal Galois field detecting the event*: it has degree $2\ell(\ell-1)$, Galois group $\mathbb{F}_\ell \rtimes (C_2 \times \mathbb{F}_\ell^\times)$ with action $(c^e, a) \cdot w = (-1)^e a w$, and $h = (c, -1)$ is a *central singleton* conjugacy class (the two signs cancel on the Kummer kernel; $g_w^2 = \tau_{2w}$ fixes the radical iff $w = 0$). Minimality is strict: since the event is the singleton $\{h\}$, no proper Galois quotient can detect it exactly; the preimage of the image of h would contain the whole coset hN. The family satisfies fiber-product disjointness by the commutator argument. *(ii) GRH range.* The discriminants obey $\log|D_{M_\ell}| < \frac{11}{2}\ell^2 \log \ell$ (tower argument; $v_5(D) = \ell(\ell-1)$ exact; $D_{M_3} = 3^{14}5^6$). The class-sensitive effective Chebotarev of Grenié–Molteni [28], applied to the singleton class ($\delta_\ell n_\ell = |C_\ell| = 1$), gives $|\pi_{E_\ell}(X) - \delta_\ell \operatorname{Li}(X)| \le 7\sqrt{X}\log X$ with *no* factor of ℓ, whence the aggregate bound $\sum_{\ell \le \sqrt{X}/(\log X)^{1+\eta}} |\pi_{E_\ell}(X) - \delta_\ell \operatorname{Li}(X)| \ll_\eta X/(\log X)^{1+\eta} = o(\pi(X))$ for every $\eta > 0$. *(iii) Logarithmic band.* For $\sqrt{X}/(\log X)^{1+\eta} < \ell \le \sqrt{X}(\log X)^\beta$, Brun–Titchmarsh on $p \equiv -1 \pmod{\ell}$ alone suffices: both endpoints have the same exponent $\frac{1}{2}$, so the Mertens mass of the band is $O(\log\log X/\log X)$ and the count is $O(X \log\log X/(\log X)^2) = o(\pi(X))$; this does not contradict the failure of the same tool on ranges with distinct exponents. *(iv) Unconditional tail.* For $\ell > \sqrt{X}(\log X)^\beta$, the quadratic trap $p^2 \mid N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(\varepsilon^{2m} - 1)$, $m = (p+1)/\ell$, $|N| < 4\varepsilon^{2m}$, gives $O_\beta(X/(\log X)^{1+2\beta})$. Combining (ii)–(iv), the uniform tail vanishes; with Theorem 12.3 and the unconditional finite products, the union passes to the infinite limit. ☐

The value 0.21272455 quoted earlier in the campaign was the truncation of the product at $\ell \le 10^6$, within its declared error; the stabilized value follows from Artin's constant via $\prod_{\ell \ne 2,5} = \mathcal{A}/\big((1-\frac{1}{2})(1-\frac{1}{20})\big)$. Against the census, $45/221 = 0.2036$ sits at -0.33 standard deviations (descriptive scale). Note the conceptual closure: *the same golden unit ε that carries the moment M_2 is the second Kummer direction of the division fields governing the tail anomalies.* Effective forms of Chebotarev [25] make the densities explicit. The depth refinements, finite unions and disjointness across primes are unconditional; the passage to the infinite union is proved here under GRH.

Two logically independent walls remain.

Wall I, H4. The equivalent forms are exclusion of local order 4, absence of split prime factors from all H_n, and saturation of the 2-part of $\mathrm{ord}_p(5)$ in the simultaneous negative-semiperiod profile. Candidates have unconditional density zero and satisfy Theorem 5.15 under GRH, but emptiness is open.

Wall II, the global union of fibers. Each fiber is finite and enumerable, seven are certified empty, and Theorem 12.1 saves one factor of $\log\log X$ for every fixed number of factors. Compatible prefixes nevertheless exist at every length by Theorem 10.15. The test case $(13, 37, 30517)$ already passes two rows, so pairwise incompatibility is false. What remains is the emptiness, or construction of a point, in the union over all compatible tuples.

A separate strengthening is to remove GRH from Theorems 5.15 and 12.4, in the spirit of unconditional partial results for Artin-type problems. Beyond $r = 5$: the moment theory of Section 3 applies verbatim to every r; only the golden identification is special to $r = 5$.

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